

Mediwatch Capstone Project

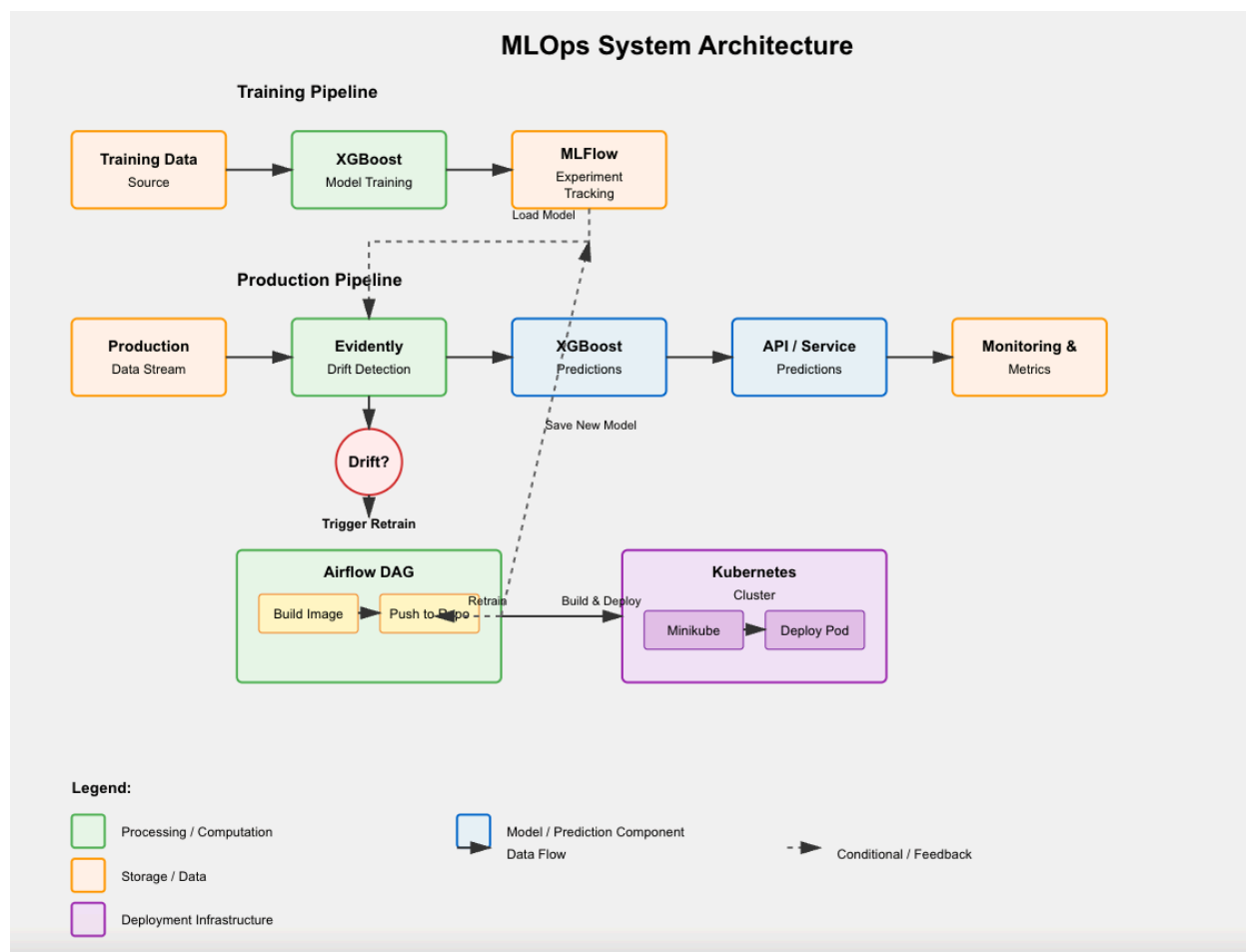
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Video Overview of my project:

<https://youtu.be/G9odoKjp2m4>

Overall diagram



Tools required

I used Ubuntu 24.04 on a local Desktop machine for this project. No CUDA is needed for this project.

You will need the following tools to run this application

Minikube, Docker and kubectl

These can be installed as follows

<https://phoenixnap.com/kb/install-minikube-on-ubuntu>

Python3

Follow the instructions below to get your system set up with python dependencies:

[Project creation](#)

[Installing requirements](#)

[PIP](#)

Local Startup

You will need to start the following if you want to run locally

MLFLOW

To do ANY training, mlflow **has** to be running. [See here](#)

Mediwatch Client App

To run the client App and perform Train and build, you need to run locally

[This is how to do it.](#)

Ports

List of ports used when running locally

PROBLEMS 12 OUTPUT DEBUG CONSOLE TERMINAL PORTS 6			
	Port	Forwarded Address	Running Process
○	5000	localhost:5000	
○	8000	127.0.0.1:8001	
○	8080	localhost:8080	
○	8443	localhost:8443	
○	9000	localhost:9000	

EDA and Data Preparation

I mostly did EDA in my FE notebook ([Mediwatch_FE.ipynb](#))

Feature Engineering

This is found in [original_base_path_training/notebooks/Mediwatch_FE.ipynb](#)

I will summarize the high points from the Feature Engineering file here.

Dropped columns

I dropped some columns because they mostly had zeros and didn't seem to contribute much. Screenshots from the notebook follow below:

▼ Drop useless encounter_id, patient_nbr

```
[ ] 1 # Drop unneeded id related columns
    2 df.drop(['encounter_id', 'patient_nbr'], axis=1, inplace=True)
```

▼ An occurrence of '?' is basically the same as a Nan. Lets update the df then do a zeros check ...

```
[ ] 1 # replace the '?'
    2 df = df.replace(r'\?', np.nan, regex=True)
    3
    4 #zeros check
    5 missing=df.isnull().sum()
    6 print(missing[missing>0].sort_values(ascending=False))
```

```
weight      98569
max_glu_serum 96420
A1Cresult    84748
medical_specialty 49949
payer_code   40256
race         2273
diag_3       1423
diag_2        358
diag_1        21
dtype: int64
```

- ✓ I will drop the columns **weight**, **max_glu_serum**, **A1Cresult**, **medical_specialty**, **payer_code**. These are about ~ 50 % of the dataset.

```
[ ] 1 # Drop unneeded columns because of NaN
2 df.drop(['weight', 'max_glu_serum', 'A1Cresult', 'medical_specialty', 'payer_code'], axis=1, inplace=True)
3
4 #zeros check
5 missing=df.isnull().sum()
6 print(missing[missing>0].sort_values(ascending=False))
7
8
```

```
race      2273
diag_3    1423
diag_2     358
diag_1      21
dtype: int64
```

Age Ranges

I decided to replace the age range with an integer value in the middle of the range

- ✓ Deal with the age column

```
[ ] 1 df['age'].unique()
```

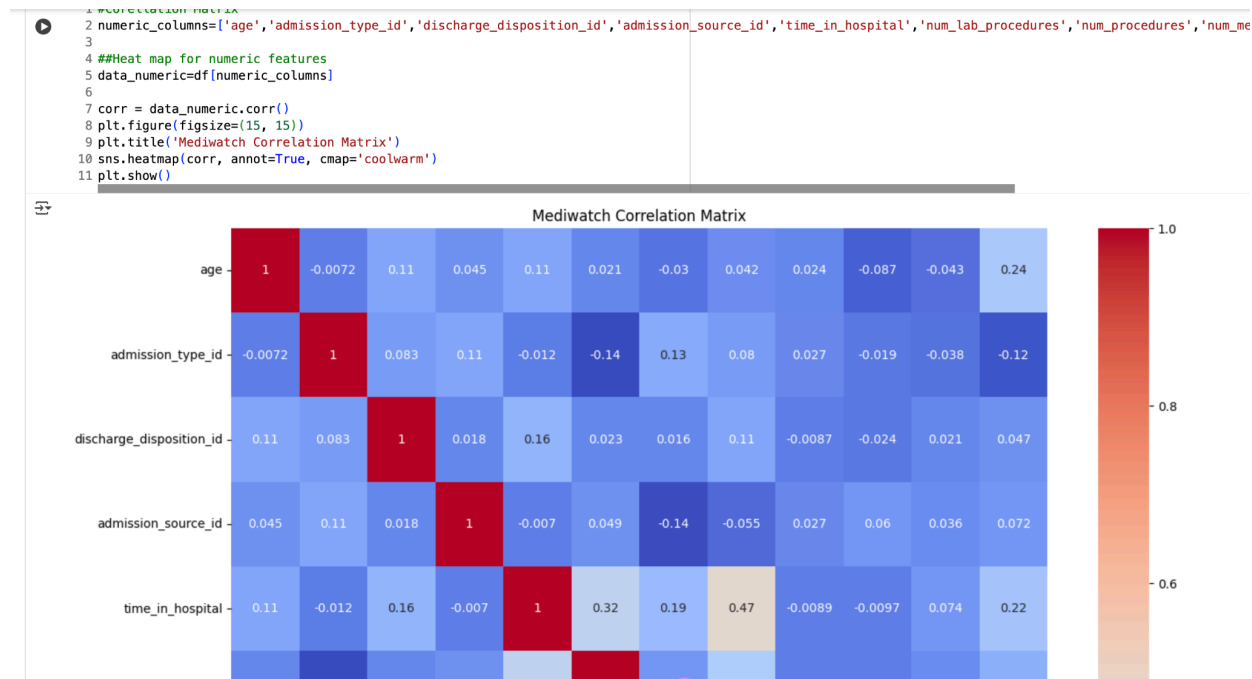
```
array(['[0-10)', '[10-20)', '[20-30)', '[30-40)', '[40-50)', '[50-60)',
      '[60-70)', '[70-80)', '[80-90)', '[90-100)'], dtype=object)
```

I will replace all ages with an integer in the middle of the row, eg 10-20 ==> 15 and so on

```
[ ] 1 df = df.replace(r'[0-10)', 5, regex=False)
2 df = df.replace(r'[10-20)', 15, regex=False)
3 df = df.replace(r'[20-30)', 25, regex=False)
4 df = df.replace(r'[30-40)', 35, regex=False)
5 df = df.replace(r'[40-50)', 45, regex=False)
6 df = df.replace(r'[50-60)', 55, regex=False)
7 df = df.replace(r'[60-70)', 65, regex=False)
8 df = df.replace(r'[70-80)', 75, regex=False)
9 df = df.replace(r'[80-90)', 85, regex=False)
10 df = df.replace(r'[90-100)', 95, regex=False)
11 df['age'].unique()
12 #df.head()
```

```
/tmp/ipython-input-40-3510274752.py:10: FutureWarning: Downcasting behavior in `replace`
df = df.replace(r'[90-100)', 95, regex=False)
array([ 5, 15, 25, 35, 45, 55, 65, 75, 85, 95])
```

Correlation Matrix



I performed a correlation matrix for numeric features. Correlation seemed pretty low.

I also did a pair plot for numeric features

Unique distribution per column

I took a look at the unique distribution per column and discovered this:

```

3
4 display_data_non_numeric=df[non_numeric_c
5
6 display_data_non_numeric.nunique()

```

	0
race	5
gender	3
diag_1	716
diag_2	748
diag_3	789
metformin	4
repaglinide	4
nateglinide	4

diag_1, diag_2, diag_3 Have very many options and can't be one hot encoded (dimensionality explosion). I will use binary encoding there instead.

Feature Engineering summary

- Some columns seem to contain '?'. These are equivalent to 'Nan' and need to be replaced with Nan
- The columns **weight, max_glu_serum, A1Cresult, medical_specialty, payer_code** need to be dropped
- The rows with 'Nan' for the columns **race, diag_3, diag_2, diag_1** need to be removed.
- Drop useless low information columns **encounter_id, patient_nbr**
- **diag_1, diag_2, diag_3** Have very many options and should not be one hot encoded (dimensionality explosion). I will binary encode these ones.
- I will replace all ages with an integer in the middle of the row, eg string '[10-20)' ==> 15 and so on

Getting Started with the Project

Project creation

I used conda to create the project initially

```
conda create -p ./env python=3.10
```

Installing requirements

Note the requirements.txt for the overall project

```
≡ requirements.txt
1  #numpy
2  numpy==1.26.4
3  #torch
4  torch==2.7.1
5  #fastapi
6  fastapi==0.116.1
7  #uvicorn
8  uvicorn==0.35.0
9  #scikit-learn
10 scikit-learn==1.7.1
11 #pandas
12 pandas==1.5.1
13 #category_encoders
14 category_encoders==2.8.1
15 #xgboost
16 xgboost==3.0.2
17 #ray[tune]
18 ray[tune]== 2.48.0
19 #mlflow
20 mlflow==3.2.0
21 #evidently
22 evidently==0.7.14
23 #jinja2
24 jinja2==3.1.4
```

Pip

Then use pip to install project requirements

```
pip install -r requirements.txt
```

Activate env

After project requirements are installed, all you need to do is activate the project like this:

```
conda activate ./env/
```

```

(base) victor@victor-Asus-PC:~/Documents/interview_kickstart/Capstone/Capstone-MLOPS-Mediwatch$
(base) victor@victor-Asus-PC:~/Documents/interview_kickstart/Capstone/Capstone-MLOPS-Mediwatch$
(base) victor@victor-Asus-PC:~/Documents/interview_kickstart/Capstone/Capstone-MLOPS-Mediwatch$
(base) victor@victor-Asus-PC:~/Documents/interview_kickstart/Capstone/Capstone-MLOPS-Mediwatch$ conda activate ./env/
/home/victor/Documents/interview_kickstart/Capstone/Capstone-MLOPS-Mediwatch/env) victor@victor-Asus-PC:~/Documents/interview_kickstart/Capstone/Capstone-MLOPS-Mediwatch$
/home/victor/Documents/interview_kickstart/Capstone/Capstone-MLOPS-Mediwatch/env) victor@victor-Asus-PC:~/Documents/interview_kickstart/Capstone/Capstone-MLOPS-Mediwatch$

```

Mediwatch UI

This project comes with a UI. I will start with a basic introduction here and then go into detail when we cover the related backend portion

To start the UI

MLFLOW

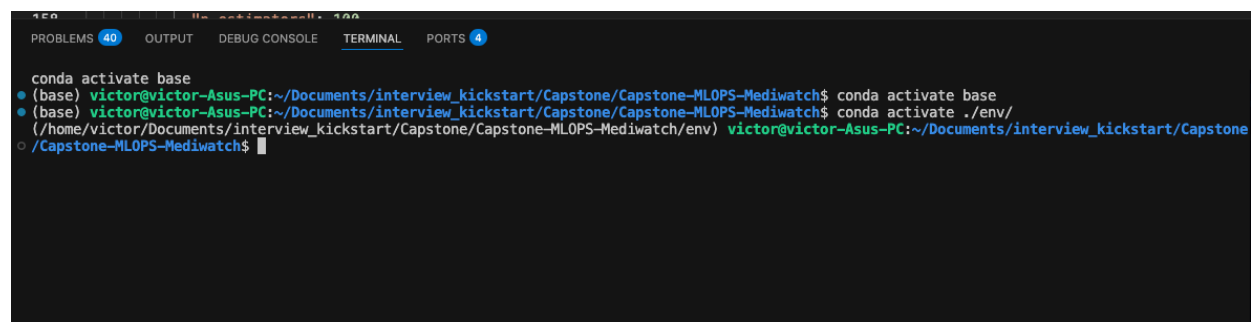
First Make sure Mlflow is running

```
mlflow ui --backend-store-uri sqlite:///mlruns.db --host 0.0.0.0 --port 5000
```

Run the Mediwatch UI

Then you can open a new terminal and activate the env

```
conda activate ./env/
```



```

150 151 152 153 154 155 156 157 158 159 160 161 162 163 164 165 166 167 168 169 170 171 172 173 174 175 176 177 178 179 180 181 182 183 184 185 186 187 188 189 190 191 192 193 194 195 196 197 198 199 200 201 202 203 204 205 206 207 208 209 210 211 212 213 214 215 216 217 218 219 220 221 222 223 224 225 226 227 228 229 230 231 232 233 234 235 236 237 238 239 240 241 242 243 244 245 246 247 248 249 250 251 252 253 254 255 256 257 258 259 260 261 262 263 264 265 266 267 268 269 270 271 272 273 274 275 276 277 278 279 280 281 282 283 284 285 286 287 288 289 290 291 292 293 294 295 296 297 298 299 300 301 302 303 304 305 306 307 308 309 310 311 312 313 314 315 316 317 318 319 320 321 322 323 324 325 326 327 328 329 330 331 332 333 334 335 336 337 338 339 340 341 342 343 344 345 346 347 348 349 350 351 352 353 354 355 356 357 358 359 360 361 362 363 364 365 366 367 368 369 370 371 372 373 374 375 376 377 378 379 380 381 382 383 384 385 386 387 388 389 390 391 392 393 394 395 396 397 398 399 400 401 402 403 404 405 406 407 408 409 410 411 412 413 414 415 416 417 418 419 420 421 422 423 424 425 426 427 428 429 430 431 432 433 434 435 436 437 438 439 440 441 442 443 444 445 446 447 448 449 450 451 452 453 454 455 456 457 458 459 460 461 462 463 464 465 466 467 468 469 470 471 472 473 474 475 476 477 478 479 480 481 482 483 484 485 486 487 488 489 490 491 492 493 494 495 496 497 498 499 500 501 502 503 504 505 506 507 508 509 510 511 512 513 514 515 516 517 518 519 520 521 522 523 524 525 526 527 528 529 530 531 532 533 534 535 536 537 538 539 540 541 542 543 544 545 546 547 548 549 550 551 552 553 554 555 556 557 558 559 560 561 562 563 564 565 566 567 568 569 570 571 572 573 574 575 576 577 578 579 580 581 582 583 584 585 586 587 588 589 590 591 592 593 594 595 596 597 598 599 600 601 602 603 604 605 606 607 608 609 610 611 612 613 614 615 616 617 618 619 620 621 622 623 624 625 626 627 628 629 630 631 632 633 634 635 636 637 638 639 640 641 642 643 644 645 646 647 648 649 650 651 652 653 654 655 656 657 658 659 660 661 662 663 664 665 666 667 668 669 670 671 672 673 674 675 676 677 678 679 680 681 682 683 684 685 686 687 688 689 690 691 692 693 694 695 696 697 698 699 700 701 702 703 704 705 706 707 708 709 710 711 712 713 714 715 716 717 718 719 720 721 722 723 724 725 726 727 728 729 730 731 732 733 734 735 736 737 738 739 740 741 742 743 744 745 746 747 748 749 750 751 752 753 754 755 756 757 758 759 760 761 762 763 764 765 766 767 768 769 770 771 772 773 774 775 776 777 778 779 780 781 782 783 784 785 786 787 788 789 790 791 792 793 794 795 796 797 798 799 800 801 802 803 804 805 806 807 808 809 810 811 812 813 814 815 816 817 818 819 820 821 822 823 824 825 826 827 828 829 830 831 832 833 834 835 836 837 838 839 840 841 842 843 844 845 846 847 848 849 850 851 852 853 854 855 856 857 858 859 860 861 862 863 864 865 866 867 868 869 870 871 872 873 874 875 876 877 878 879 880 881 882 883 884 885 886 887 888 889 890 891 892 893 894 895 896 897 898 899 900 901 902 903 904 905 906 907 908 909 910 911 912 913 914 915 916 917 918 919 920 921 922 923 924 925 926 927 928 929 930 931 932 933 934 935 936 937 938 939 940 941 942 943 944 945 946 947 948 949 950 951 952 953 954 955 956 957 958 959 960 961 962 963 964 965 966 967 968 969 970 971 972 973 974 975 976 977 978 979 980 981 982 983 984 985 986 987 988 989 990 991 992 993 994 995 996 997 998 999 1000
PROBLEMS (40) OUTPUT DEBUG CONSOLE TERMINAL PORTS (4)
conda activate base
(base) victor@victor-Asus-PC:~/Documents/interview_kickstart/Capstone/Capstone-MLOPS-Mediwatch$ conda activate base
(base) victor@victor-Asus-PC:~/Documents/interview_kickstart/Capstone/Capstone-MLOPS-Mediwatch$ conda activate ./env/
(/home/victor/Documents/interview_kickstart/Capstone/Capstone-MLOPS-Mediwatch/env) victor@victor-Asus-PC:~/Documents/interview_kickstart/Capstone/Capstone-MLOPS-Mediwatch$

```

Go to client_app folder

```
cd client_app/
```

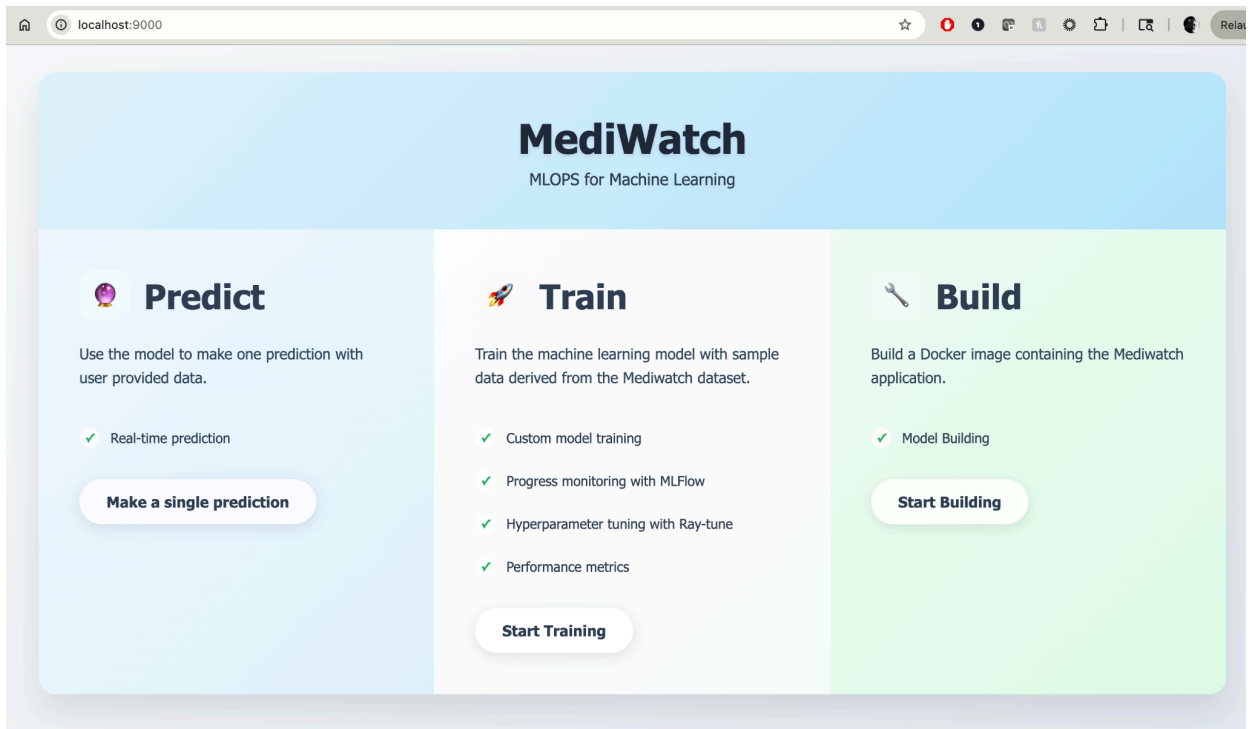
Run the Client App

```
python3 ClientApp.py
```

```
drwxrwxr-x 2 victor victor 4096 Sep 1 11:27 templates
(/home/victor/Documents/interview_kickstart/Capstone/Capstone-MLOPS-Mediwa
o /Capstone-MLOPS-Mediwatch/client_app$ python3 ClientApp.py
INFO:      Started server process [3467489]
INFO:      Waiting for application startup.
INFO:      Application startup complete.
INFO:      Uvicorn running on http://0.0.0.0:9000 (Press CTRL+C to quit)
INFO:      127.0.0.1:59614 - "GET / HTTP/1.1" 200 OK
```

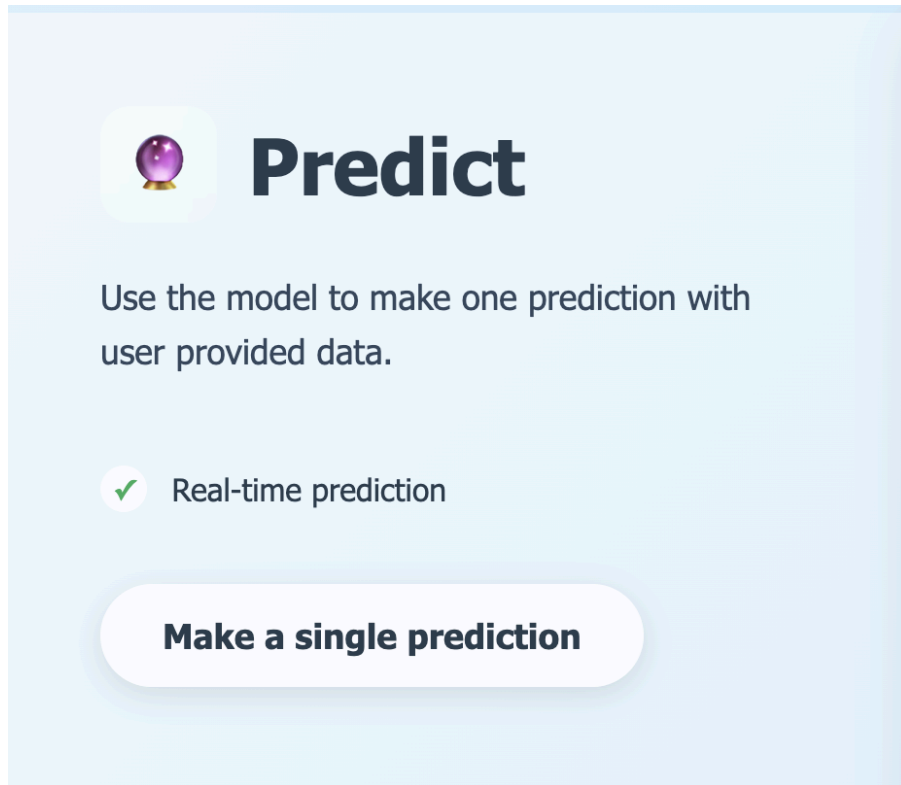
UI Frontend

This is the UI that will run on port 9000



Predict

Make one prediction with manually entered data and return data to the frontend. This covers inference and will be detailed later on



Train

This triggers hyperparameter tuning with **ray-tune**. Many experiments can be run and the results are available in MIFlow where many experiments can be run and the results viewed.



Train

Train the machine learning model with sample data derived from the Mediwatch dataset.

- ✓ Custom model training
- ✓ Progress monitoring with MLFlow
- ✓ Hyperparameter tuning with Ray-tune
- ✓ Performance metrics

Start Training

Build

This will Build a Docker image containing a working application



Build

Build a Docker image containing the Mediwatch application.



Model Building

Start Building

Mediwatch UI Code

FastAPI documentation

This is available at /docs endpoint. A screenshot of the FastAPI endpoints are shown below



FastAPI 0.1.0 OAS 3.1

/openapi.json

default

GET	/	Home
GET	/health	Health
GET	/single_prediction	Single Prediction Page
POST	/single_prediction	Single Prediction Post
GET	/training	Training
GET	/start_training	Start Training
GET	/building	Building
POST	/dockerize	Dockerize
GET	/dockerize_stream	Dockerize Stream

The actual client code is found at **client_app/ClientApp.py**

This has all the endpoints described above. We will go through all of them in later in the document.

Training (basic)

I found it useful to create the **original_base_path_training/notebooks/Mediwatch_train.ipynb** as the starting point for training my model.

The actual training in the project is done by the **original_base_path_training/[training.py](#)** file.

```

/home/victor/Documents/interview_kickstart/Capstone/Capstone-MLOPS-Mediwa
(/home/victor/Documents/interview_kickstart/Capstone/Capstone-MLOPS-Mediwa
-Mediwatch/original_base_path_training$ ls -l
total 18764
-rw-rw-r-- 1 victor victor    25441 Sep 29 22:36 confusion_matrix.png
-rw-rw-r-- 1 victor victor 19159383 Sep 29 22:36 diabetic_data.csv
-rw-rw-r-- 1 victor victor      0 Aug  2 23:12 __init__.py
drwxrwxr-x 2 victor victor    4096 Jul 30 22:03 notebooks
drwxrwxr-x 2 victor victor    4096 Sep 30 21:48 pycache
-rw-rw-r-- 1 victor victor    12830 Sep 30 21:43 training.py
(/home/victor/Documents/interview_kickstart/Capstone/Capstone-MLOPS-Mediwa
-Mediwatch/original_base_path_training$

```

Verify and clean up data

Look at the

```
process_mediwatch_dataset(df):
```

This function systematically cleans up incoming data

- 1) Verify existence of columns

```

for col in required_cols:
    assert col in df.columns, f"{col} is not in the provided dataframe"

```

- 2) Map the target variable

```

# deal with target variable
df['readmitted'] = df['readmitted'].map({'NO': 0, '<30': 1, '>30': 2 })

```

- 3) Replace nans and drop unneeded columns

```

# replace '?' in several places in the df with Nan
df = df.replace(r'\?', np.nan, regex=True)

# Drop unneeded columns because of NaN
df.drop(['weight', 'max_glu_serum', 'A1Cresult', 'medical_specialty', 'payer_code'], axis=1, inplace=True)

# Drop remaining rows with missing values
df.dropna(inplace=True)

# Drop unneeded id related columns
df.drop(['encounter_id', 'patient_nbr'], axis=1, inplace=True)

```

- 4) Deal with age column

```
#Deal with the age column
df = df.replace(r'[0-10]', 5, regex=False)
df = df.replace(r'[10-20]', 15, regex=False)
df = df.replace(r'[20-30]', 25, regex=False)
df = df.replace(r'[30-40]', 35, regex=False)
df = df.replace(r'[40-50]', 45, regex=False)
df = df.replace(r'[50-60]', 55, regex=False)
df = df.replace(r'[60-70]', 65, regex=False)
df = df.replace(r'[70-80]', 75, regex=False)
df = df.replace(r'[80-90]', 85, regex=False)
df = df.replace(r'[90-100]', 95, regex=False)
```

Vectorize Datasets

Take a look at the

```
def vectorize_datasets(X_train, X_test):
```

Here I perform one hot encoding on most of the columns

```
#one hot encode the following columns: ['race','gender','metformin','repaglinide','nateglinide','chlorpropamide','glimepiride',
cols_to_one_hot_encode= ['race','gender','metformin','repaglinide','nateglinide','chlorpropamide','glimepiride','acetoexamide',

# One hot encoder
one_hot_encoder = OneHotEncoder(handle_unknown='ignore',sparse_output=False, drop='first')

# Fit to only training data
one_hot_encoder.fit(X_train[cols_to_one_hot_encode])

#Transform Test/Train data
X_train_transformed_one_hot = one_hot_encoder.transform(X_train[cols_to_one_hot_encode])
X_test_transformed_one_hot = one_hot_encoder.transform(X_test[cols_to_one_hot_encode])

# Create Dataframe with encoded columns
feature_names = one_hot_encoder.get_feature_names_out(cols_to_one_hot_encode)
#encoded_df = pd.DataFrame(encoded_array, columns=feature_names, index=df.index)
X_train_encoded_one_hot = pd.DataFrame(X_train_transformed_one_hot, columns=feature_names, index=X_train.index)
X_test_encoded_one_hot = pd.DataFrame(X_test_transformed_one_hot, columns=feature_names, index=X_test.index)

# Combine with remaining columns
#df = pd.concat([df.drop(cols_to_one_hot_encode, axis=1), encoded_df], axis=1)
X_train = pd.concat([X_train.drop(cols_to_one_hot_encode, axis=1), X_train_encoded_one_hot], axis=1)
X_test = pd.concat([X_test.drop(cols_to_one_hot_encode, axis=1), X_test_encoded_one_hot], axis=1)
```

However for the 'diag_1','diag_2','diag_3' identified in the FE, I perform binary encoding

```

#binary encode ['diag_1','diag_2','diag_3']
binary_encoder = BinaryEncoder(cols=['diag_1','diag_2','diag_3'])
# Fit and transform specific columns
#df = binary_encoder.fit_transform(df)
cols_to_binary_encode = ['diag_1','diag_2','diag_3']

# Fit to only training data
binary_encoder.fit(X_train[cols_to_binary_encode])

#Transform Test/Train data
X_train_transformed_binary = binary_encoder.transform(X_train[cols_to_binary_encode])
X_test_transformed_binary = binary_encoder.transform(X_test[cols_to_binary_encode])

# Create Dataframe with encoded columns
feature_names = binary_encoder.get_feature_names_out(cols_to_binary_encode)
#encoded_df = pd.DataFrame(encoded_array, columns=feature_names, index=df.index)
X_train_encoded_binary = pd.DataFrame(X_train_transformed_binary, columns=feature_names, index=X_train.index)
X_test_encoded_binary = pd.DataFrame(X_test_transformed_binary, columns=feature_names, index=X_test.index)

# Combine with remaining columns
X_train = pd.concat([X_train.drop(cols_to_binary_encode, axis=1), X_train_encoded_binary], axis=1)
X_test = pd.concat([X_test.drop(cols_to_binary_encode, axis=1), X_test_encoded_binary], axis=1)

```

Model Export

I export the model and the encoders as pickle files

```

def export_model_to_disk(model, one_hot_encoder, binary_encoder, data_directory):
    #model
    model_path = os.path.join(data_directory, MODEL_PICKLE_FILE)
    with open(model_path, 'wb') as file:
        pickle.dump(model, file)

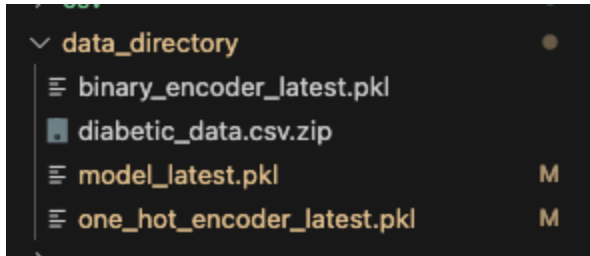
    #one hot encoder
    one_hot_encoder_path = os.path.join(data_directory, ONE_HOT_ENCODER_PICKLE_FILE)
    with open(one_hot_encoder_path, 'wb') as file:
        pickle.dump(one_hot_encoder, file)

    #Binary encoder
    binary_encoder_path = os.path.join(data_directory, BINARY_ENCODER_PICKLE_FILE)
    with open(binary_encoder_path, 'wb') as file:
        pickle.dump(binary_encoder, file)

```

I chose pickle because it is a standard python format that is easily portable.

These files are exported to the **data_directory**



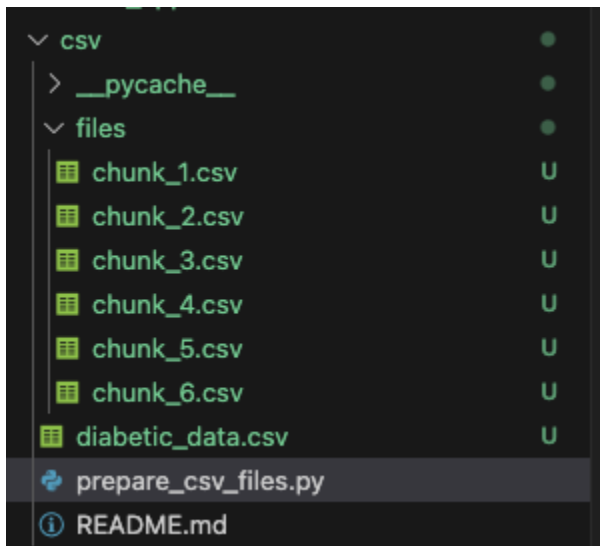
Divide up the original Dataset

Take a look at the

```
csv/prepare_csv_files.py
```

I divide the large dataset provided for the project into many small subsets (chunks) to use for training and prediction.

This is a simple way to use one chunk to approximate initial data and then later chunks to approximate later data.



Manually Run training (basic)

Start Mlflow first

```
mlflow ui --backend-store-uri sqlite:///mlruns.db --host 0.0.0.0 --port 5000
```

```
python3 training.py
```

```

● -Mediawatch/original_base_path_training$ python3 training.py
Preparing data...
File chunk_1.csv is removed
File chunk_2.csv is removed
File chunk_4.csv is removed
File chunk_6.csv is removed
File chunk_5.csv is removed
File chunk_3.csv is removed
The csv file extracted is diabetic_data.csv
The rows in csv are 101766 , the chunk size is 4000 ,the number of csv file produced are 6
The csv file obtained is ../csv/files/chunk_1.csv
XGBoost Accuracy: 0.1540880503144654
XGBoost Recall (weighted): 0.1540880503144654
Class 1 (Under 30 day readmission) Recall: 0.9745222929936306
Class 1 (Under 30 day readmission) F1-Score: 0.20393202265911362
      precision    recall  f1-score   support

Class 0         0.78         0.09         0.16         1431
Class 1         0.11         0.97         0.20          314
Class 2         0.64         0.01         0.01         1117

 accuracy         0.15         0.15         0.15         2862
 macro avg        0.51         0.36         0.13         2862
 weighted avg     0.65         0.15         0.11         2862

🐘 View run vaunted-crab-216 at: http://localhost:5000/#/experiments/0/runs/e2df1649758d46c79b75177e1daa8221
📄 View experiment at: http://localhost:5000/#/experiments/0
Time taken (seconds): 37
(there is a link to the experiment in the bottom right corner of the page)

```

You can see the training results in Mlflow

The screenshot displays the mlflow web interface for managing experiments. The left sidebar contains navigation links for "Experiments", "Models", and "Prompts". The main area shows the "Default" experiment with options to "Provide Feedback" or "Add Description". Below this are tabs for "Runs", "Models", "Experimental", "Evaluation", and "Traces". A search bar contains the query "metrics.rmse < 1 and params.model = 'tree'", and filters show "Time created", "State: Active", and "Datasets". A sorting section indicates "Sort: Created", "Columns", and "Group by".

	Run Name	Created ↕	Dataset	Duration	Source	Models
☐	vaunted-crab-216	✓ 3 minutes ago	-	37.6s	📄 training...	-
☐	unleashed-lark-198	✓ 3 days ago	-	58.3s	📄 ray_tun...	-
☐	languid-mouse-235	✓ 3 days ago	-	1.0min	📄 ray_tun...	-

2.0

Default >

vaunted-crab-216

Overview

Model metrics

System metrics

Traces

Artifacts

Status	🟢 Finished
Run ID	e2df1649758d46c79b75177e1daa8221
Duration	37.6s
Datasets used	—
Tags	Add tags
Source	training.py 11e7c8f
Logged models	—
Registered models	—
Registered prompts	—

Metrics (4)

Metric	Value
Train Accuracy	0.1540880503144654
Train Recall weighted	0.1540880503144654
Class 1 Under 30 day readmission Recall	0.9745222929936306
Class 1 Under 30 day readmission F1-Score	0.20393202265911362

Parameters (7)

Parameter	Value
n_estimators	100
max_depth	3
learning_rate	0.01
subsample	0.6
gamma	0
reg_alpha	1e-05

Note that the training focuses on ‘**Class 1**’ which is the ‘**readmitted under 30 Days**’ both for Recall and F1 Score.

Default > vaunted-crab-216 >

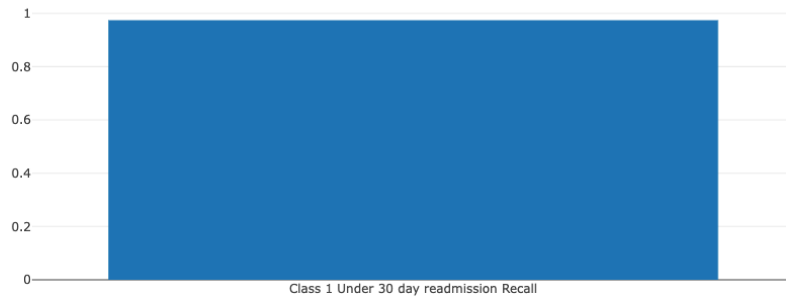
Class 1 Under 30 day readmission Recall

Y-axis:

Class 1 Under 30 day readmission Recall X

Y-axis Log Scale: ☐

Download data



Metric	Latest	Min	Max
Class 1 Under 30 day readmission Recall	0.9745222929936306 (step=0)	0.9745222929936306 (step=0)	0.9745222929936306 (step=0)

Default > vaunted-crab-216 >

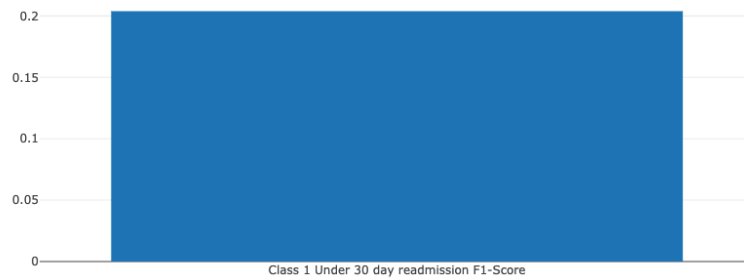
Class 1 Under 30 day readmission F1-Score

Y-axis:

Class 1 Under 30 day readmission F1-Score X

Y-axis Log Scale: ☐

Download data



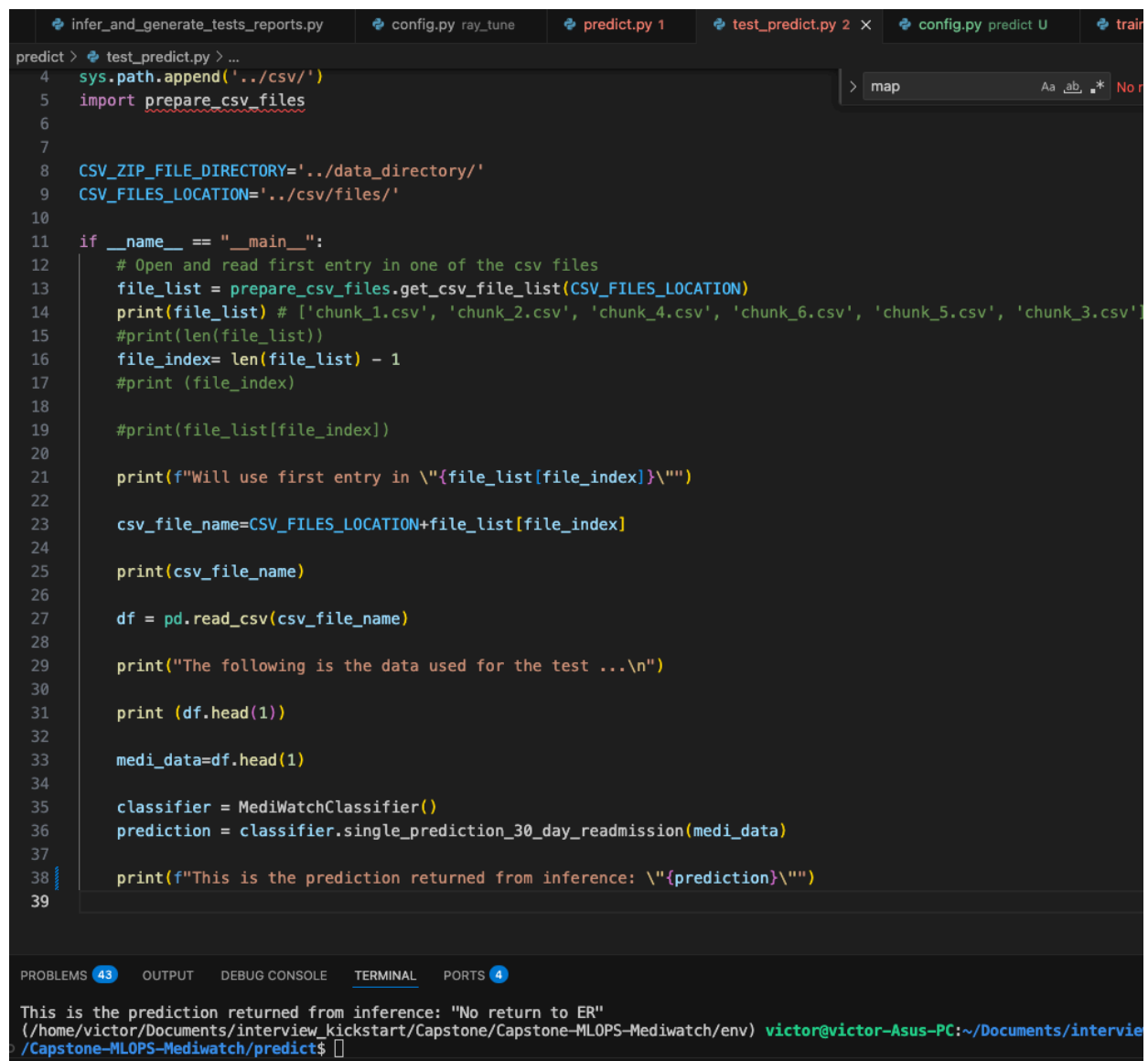
Metric	Latest	Min	Max
Class 1 Under 30 day readmission F1-Score	0.20393202265911362 (step=0)	0.20393202265911362 (step=0)	0.20393202265911362 (step=0)

Inference

I found it useful to create the `original_base_path_training/notebooks/Mediwatch_infer.ipynb` as the starting point for inference.

The actual inference in the project is done by the `predict/predict.py` file.

I used the `predict/test_predict.py` to check basic operation



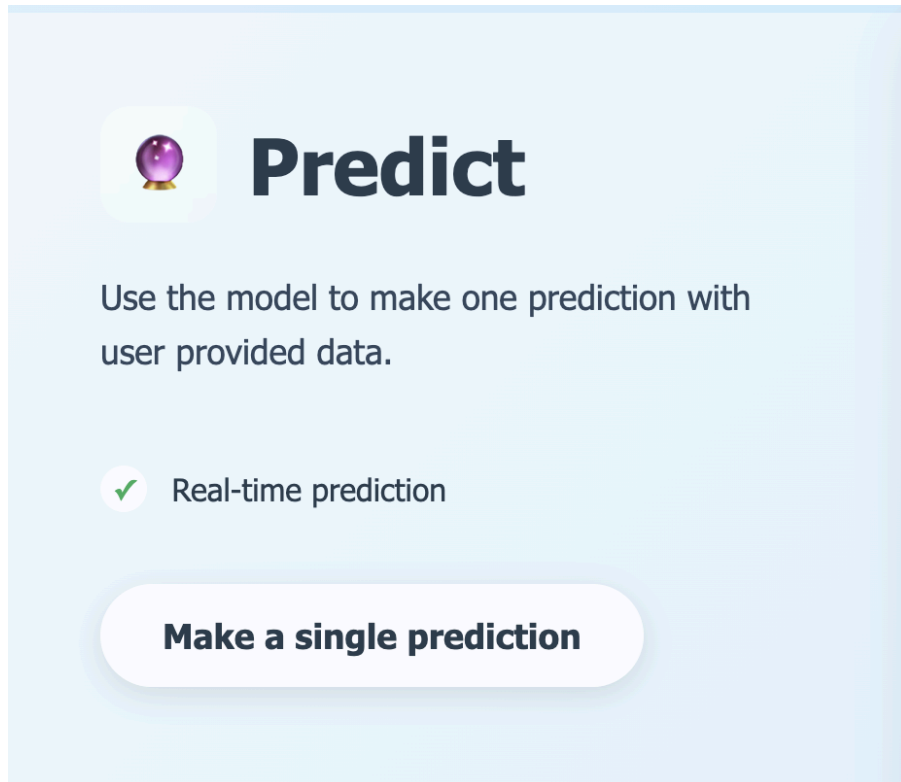
```
predict > test_predict.py > ...
4 sys.path.append('../csv/')
5 import prepare_csv_files
6
7
8 CSV_FILE_DIRECTORY='../data_directory/'
9 CSV_FILES_LOCATION='../csv/files/'
10
11 if __name__ == "__main__":
12     # Open and read first entry in one of the csv files
13     file_list = prepare_csv_files.get_csv_file_list(CSV_FILES_LOCATION)
14     print(file_list) # ['chunk_1.csv', 'chunk_2.csv', 'chunk_4.csv', 'chunk_6.csv', 'chunk_5.csv', 'chunk_3.csv']
15     #print(len(file_list))
16     file_index= len(file_list) - 1
17     #print (file_index)
18
19     #print(file_list[file_index])
20
21     print(f"Will use first entry in \"{file_list[file_index]}\"")
22
23     csv_file_name=CSV_FILES_LOCATION+file_list[file_index]
24
25     print(csv_file_name)
26
27     df = pd.read_csv(csv_file_name)
28
29     print("The following is the data used for the test ...\n")
30
31     print (df.head(1))
32
33     medi_data=df.head(1)
34
35     classifier = MediWatchClassifier()
36     prediction = classifier.single_prediction_30_day_readmission(medi_data)
37
38     print(f"This is the prediction returned from inference: \"{prediction}\"")
39
```

PROBLEMS 43 OUTPUT DEBUG CONSOLE TERMINAL PORTS 4

This is the prediction returned from inference: "No return to ER"
(/home/victor/Documents/interview_kickstart/Capstone/Capstone-MLOPS-Mediwatch/env) victor@victor-Asus-PC:~/Documents/interview_kickstart/Capstone-MLOPS-Mediwatch/predict\$

UI Front End

Predict is actually used by the front end to run the model and return quick predictions to the end user:



After you Click on the 'Make a Single prediction' control, you get a form where you can enter all the information for your subject. Note that several fields are pre-populated with some default values.



Mediwatch Single Prediction Form

PATIENT INFORMATION

Race

Select Race



Gender

Select Gender



Age *

Select Age Range



Weight

ADMISSION INFORMATION

Admission Type ID

0

Discharge Disposition ID

0

Admission Source ID

0

Time in Hospital (days)

0

Payer Code

None

Medical Specialty

...

...

COMBINATION MEDICATIONS

Glyburide-Metformin *

No

Glipizide-Metformin *

No

Glimepiride-Pioglitazone *

No

Metformin-Rosiglitazone *

No

Metformin-Pioglitazone *

No

ADDITIONAL INFORMATION

Change in Medication

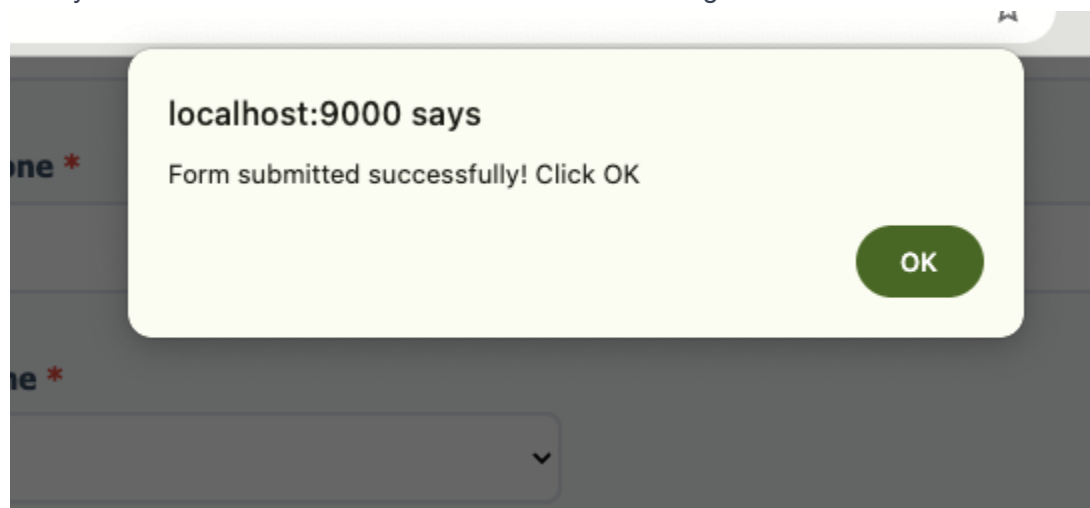
Select

Diabetes Medication Prescribed

Select

MAKE SINGLE PREDICTION

After you enter the client information and click 'Make Single Prediction'



Prediction results are shown on the screen

MAKE SINGLE PREDICTION

PREDICTION RESULTS

No return to ER

This endpoint calls the `single_prediction` endpoint (see FastAPI [here](#))

POST **/single_prediction** Single Prediction Post

Parameters

No parameters

Request body required

[Example Value](#) | [Schema](#)

```
{
  "race": "string",
  "gender": "string",
  "age": "string",
  "weight": "string",
  "admission_type_id": 0,
  "discharge_disposition_id": 0,
  "admission_source_id": 0,
  "time_in_hospital": 0,
  "payer_code": "string",
  "medical_specialty": "string",
  "num_lab_procedures": 0,
  "num_procedures": 0,
  "num_medications": 0,
  "number_outpatient": 0,
  "number_emergency": 0,
  "number_inpatient": 0,
  "diag_1": "string",
  "diag_2": "string",
  "diag_3": "string",
  "number_diagnoses": 0,
  "metformin": "string",
  "repaglinide": "string",
  "nateglinide": "string",
  "chlorpropamide": "string"
}
```

And the call itself:

Call:

Name	X	Headers	Payload	Preview	Response	Initiator	Timing	Cookies
single_prediction		▼ General						
		Request URL			http://localhost:9000/single_prediction			
		Request Method			POST			
		Status Code			200 OK			
		Remote Address			127.0.0.1:9000			
		Referrer Policy			strict-origin-when-cross-origin			
		▼ Response Headers			☐ Raw			
		Access-Control-Allow-Credentials			true			
		Access-Control-Allow-Origin			http://localhost:9000			
		Content-Length			50			
		Content-Type			application/json			
		Date			Sun, 05 Oct 2025 17:25:02 GMT			
1 requests	281 B transferred	Server			uvicorn			

Response:

Name	X	Headers	Payload	Preview	Response	Initiator	Timing	Cookies
single_prediction		1			{"prediction": "No return to ER", "confidence": null}			

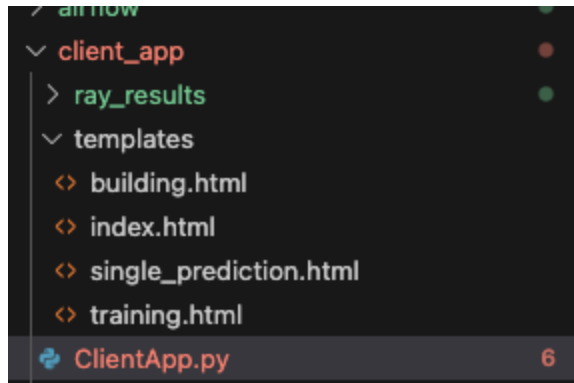
Once again, refer to:

client_app/Client_App.py

client_app/templates

Prediction Template

The single prediction template is found here:



Specifically **client_app/templates/single_prediction.html** renders the UI used for the prediction page

ClientApp.py

The actual backend code is found in **client_app/Client_App.py** under the 'single_prediction_post' function. The call to the prediction is on line 139 below


```

client_app > ClientApp.py > single_prediction_post > result
126     logs: list[str] = None
127
128 @app.post("/single_prediction", response_model=PredictionResponse)
129 async def single_prediction_post(prediction_data: MediwatchSubject):
130     medi_data=prediction_data
131
132     # convert pydantic model to a dictionary
133     medi_data_dict=medi_data.model_dump(by_alias=True)
134
135     # convert dictionary to a dataframe
136     medi_data_df=pd.DataFrame([medi_data_dict])
137
138     # send dataframe to classifier
139     result=clApp.classifier.single_prediction_30_day_readmission(medi_data_df)
140
141     # Convert numpy array to string or list
142     if isinstance(result, np.ndarray):
143         print ("result is an np.ndarray") # TODO DEBUG!!
144         prediction_value = result.tolist() # Convert to Python list
145     else:
146         prediction_value = str(result)
147
148     return PredictionResponse(
149         prediction=str(prediction_value) # Ensure it's a string
150     )
151

```

Large Scale Hyperparameter Optimization and Experiment Tracking

Ray Tune

I used ray tune to optimize hyperparameter tuning for my project. The ray tune code is located in the **ray_tune** folder.

ray_tune/ray_tune.py contains the code.

The main function in the ray_tune.py is the

```
diabetic_data_main_with_ray_tune()
```

Here is a screenshot of part of the function:

```
def diabetic_data_main_with_ray_tune():
    # initialize mlflow
    mlflow.set_tracking_uri(cfg.MLFLOW_TRACKING_URI)
    # Initialize Ray
    if not ray.is_initialized():
        ray.init()
    start_time = datetime.now()
    # Prepare data once
    print("Preparing data...")
    X_train, X_test, y_train, y_test, one_hot_encoder, binary_encoder = training.prepare_data(cfg.CSV_ZIP_FILE_DIRECTORY, cfg.CSV_ZIP_FILE_NAME)

    ## Train
    data_dict={
        "X_train":X_train,
        "X_test":X_test,
        "y_train":y_train,
        "y_test":y_test
    }

    search_space ={
        "n_estimators": tune.choice(cfg.N_ESTIMATORS),
        "max_depth": tune.choice(cfg.MAX_DEPTH),
        "learning_rate": tune.loguniform(cfg.LEARNING_RATE_LOWER, cfg.LEARNING_RATE_UPPER),
        "subsample": tune.uniform(cfg.SUBSAMPLE_LOWER, cfg.SUBSAMPLE_UPPER),
        "colsample_bytree": tune.uniform(cfg.COLSAMPLE_BYTREE_LOWER, cfg.COLSAMPLE_BYTREE_UPPER),
```

Ray Tune Parameters

ray_tune/config.py contains the parameters to tune.

This is a quick screenshot of the parameters I used in the config file

```
NUM_SAMPLES = 50
N_ESTIMATORS=[100, 200, 300, 500, 800]
MAX_DEPTH=[3, 4, 5, 6, 7, 8]           # maximum depth of the tree
LEARNING_RATE_LOWER=0.01               # learning rate (aka eta)
LEARNING_RATE_UPPER=0.3                # learning rate (aka eta)
SUBSAMPLE_LOWER=0.6
SUBSAMPLE_UPPER=1.0
COLSAMPLE_BYTREE_LOWER=0.6
COLSAMPLE_BYTREE_UPPER=1.0
GAMMA_LOWER=0
GAMMA_UPPER=5
REG_ALPHA_LOWER=1e-5
REG_ALPHA_UPPER=10
REG_LAMBDA_LOWER=1e-5
REG_LAMBDA_UPPER=10
```

Manually running ray_tune.py

Firstly activate the env

```
conda activate ./env/
```

Then run ray_tune

```
python3 ray_tune.py
```

colsample_bytree	0.85648
gamma	4.30944
learning_rate	0.25042
max_depth	6
n_estimators	200
reg_alpha	0.00309
reg_lambda	4e-05
subsample	0.94575

Trial train_xgboost_self_contained_bfe40_00020 completed after 1 iterations at 2025-10-05 20:52:51. Total running time: 13s

Trial train_xgboost_self_contained_bfe40_00020 result	
checkpoint_dir_name	
time_this_iter_s	2.79463
time_total_s	2.79463
training_iteration	1
accuracy	0.56918
class_1_recall	0.00955

Trial train_xgboost_self_contained_bfe40_00019 completed after 1 iterations at 2025-10-05 20:52:51. Total running time: 13s

Trial train_xgboost_self_contained_bfe40_00019 result	
checkpoint_dir_name	
time_this_iter_s	3.21571
time_total_s	3.21571
training_iteration	1
accuracy	0.58141
class_1_recall	0

Trial train_xgboost_self_contained_bfe40_00022 completed after 1 iterations at 2025-10-05 20:52:52. Total running time: 14s

...

...

PROBLEMS 36 OUTPUT DEBUG CONSOLE TERMINAL PORTS 4 bash - ray_tune												
pstone/Capstone-MLOPS-Mediwatch/ray_tune/ray_results/xgboost_tune' in 0.0181s.												
Trial status: 50 TERMINATED												
Current time: 2025-10-05 20:53:08. Total running time: 30s												
Logical resource usage: 1.0/12 CPUs, 0/0 GPUs												
Current best trial: bfe40_00015 with class_1_recall=0.07643312101910828 and params={'n_estimators': 100, 'max_depth': 8, 'learning_rate': 0.29364315257118473, 'subsample': 0.8496022353158019, 'colsample_bytree': 0.9496337287218778, 'gamma': 0.21862311940031243, 'reg_alpha': 7.986917103852926e-05, 'reg_lambda': 0.0006543141252716773}												
Trial name	reg_lambda	iter	total time (s)	status	accuracy	n_estimators	max_depth	learning_rate	subsample	colsample_bytree	gamma	reg_a
lpha	reg_lambda	iter	total time (s)	status	accuracy	class_1_recall						
train_xgboost_self_contained_bfe40_00000				TERMINATED		300						
164 0.00012712		1	3.0762	0.581062	0		3	0.0302248	0.868344	0.609745	1.61803	0.00366
train_xgboost_self_contained_bfe40_00001				TERMINATED		200						
2203 8.73378e-05		1	2.05784	0.579665	0		6	0.190153	0.718605	0.674733	4.96167	0.00014
train_xgboost_self_contained_bfe40_00002				TERMINATED		500						
3243 0.000575427		1	4.68148	0.577568	0.00636943		5	0.0432915	0.735021	0.65563	0.894336	0.00035
train_xgboost_self_contained_bfe40_00003				TERMINATED		500						
5.13632		1	3.56577	0.577918	0		7	0.131234	0.945586	0.673671	1.24373	7.00305
train_xgboost_self_contained_bfe40_00004				TERMINATED		500						
e-05 0.00156412		1	3.7445	0.580363	0		3	0.0355454	0.688324	0.975618	3.79403	2.61611
train_xgboost_self_contained_bfe40_00005				TERMINATED		500						
3359 0.00903305		1	4.05882	0.573725	0.0127389		7	0.107106	0.887806	0.652368	0.821343	0.00094
train_xgboost_self_contained_bfe40_00006				TERMINATED		500						
61 0.00912108		1	6.84514	0.569182	0		7	0.0286155	0.625363	0.634147	0.863229	0.00642
train_xgboost_self_contained_bfe40_00007				TERMINATED		300						
31 0.00319205		1	3.14701	0.581412	0.00318471		7	0.0407895	0.964584	0.762213	1.90452	0.02037
train_xgboost_self_contained_bfe40_00008				TERMINATED		200						
62 1.62245e-05		1	2.1459	0.581761	0		4	0.0619453	0.677971	0.698736	3.75309	0.04215
train_xgboost_self_contained_bfe40_00009				TERMINATED		100						
174 0.0198721		1	2.03801	0.576869	0		5	0.0212341	0.789488	0.972532	2.5643	0.00133
train_xgboost_self_contained_bfe40_00010				TERMINATED		200						
47 0.966877		1	2.19848	0.580713	0		3	0.0491496	0.93682	0.761843	3.1525	0.02148
train_xgboost_self_contained_bfe40_00011				TERMINATED		500						
341 1.96503e-05		1	4.92564	0.58246	0.00636943		4	0.0567145	0.697924	0.831668	0.927453	0.00154

...

...

```
Best hyperparameters found:
n_estimators: 100
max_depth: 8
learning_rate: 0.29364315257118473
subsample: 0.8496022353158019
colsample_bytree: 0.9496337287218778
gamma: 0.21862311940031243
reg_alpha: 7.986917103852926e-05
reg_lambda: 0.0006543141252716773

Best Class 1 recall: 0.0764
Best accuracy: 0.5461

Training final model with best hyperparameters...

Final Model Performance:
XGBoost Accuracy: 0.5461215932914046
XGBoost Class 1 Recall: 0.07643312101910828
precision    recall  f1-score   support

      0       0.59       0.71       0.64      1431
      1       0.32       0.08       0.12       314
      2       0.50       0.47       0.48      1117

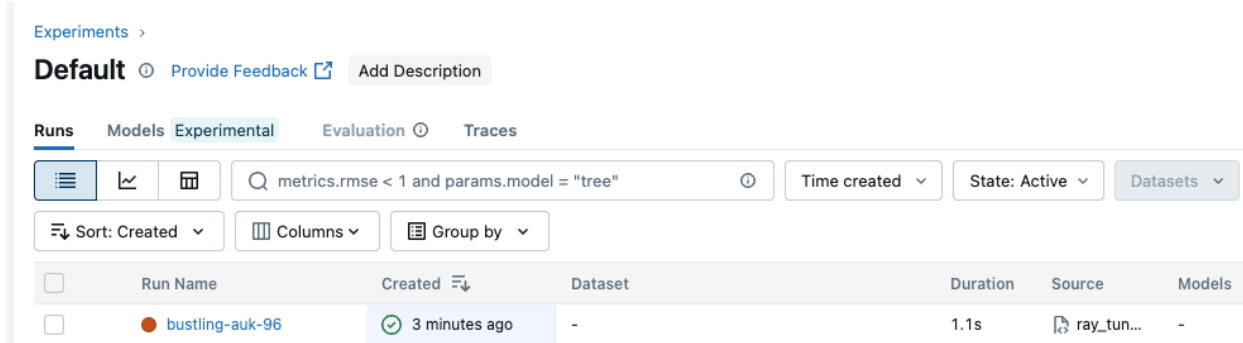
 accuracy          0.47          0.42          0.55      2862
 macro avg          0.47          0.42          0.42      2862
 weighted avg          0.52          0.55          0.52      2862

View run bustling-auk-96 at: http://localhost:5000/#/experiments/0/runs/59d67efeced349e189480c803e52cafa
View experiment at: http://localhost:5000/#/experiments/0
```

The model is finally trained with the best hyperparameters available

MLFLOW

The ray_tune run is reported in mlflow. Here is the run we just performed:



As you can see, all the ray parameters are reported:

Default >

bustling-auk-96



Overview

Model metrics

System metrics

Traces

Artifacts

Tags	Add tags
Source	ray_tune.py 1d32d50
Logged models	—
Registered models	—
Registered prompts	—

Metrics (2)

Search metrics	
Metric	Value
Train Accuracy	0.5461215932914046
Train Class 1 Recall	0.07643312101910828

Parameters (10)

Search parameters	
Parameter	Value
n_estimators	100
max_depth	8
learning_rate	0.29364315257118473
subsample	0.8496022353158019
colsample_bytree	0.9496337287218778
gamma	0.21862311940031243
reg_alpha	7.986917103852926e-05
reg_lambda	0.0006543141252716773
eval_metric	logloss
random_state	42

Default >

bustling-auk-96

Overview

Model metrics

System metrics

Traces

Artifacts

- plots
- classification_report.txt

classification_report.txt 380B

Path: mlflow-artifacts/0/59d67efeced349e189480c803e52cafa/artifacts/classification_report.txt

	precision	recall	f1-score	support
0	0.59	0.71	0.64	1431
1	0.32	0.08	0.12	314
2	0.50	0.47	0.48	1117
accuracy			0.55	2862
macro avg	0.47	0.42	0.42	2862
weighted avg	0.52	0.55	0.52	2862

Default >

bustling-auk-96

Overview

Model metrics

System metrics

Traces

Artifacts

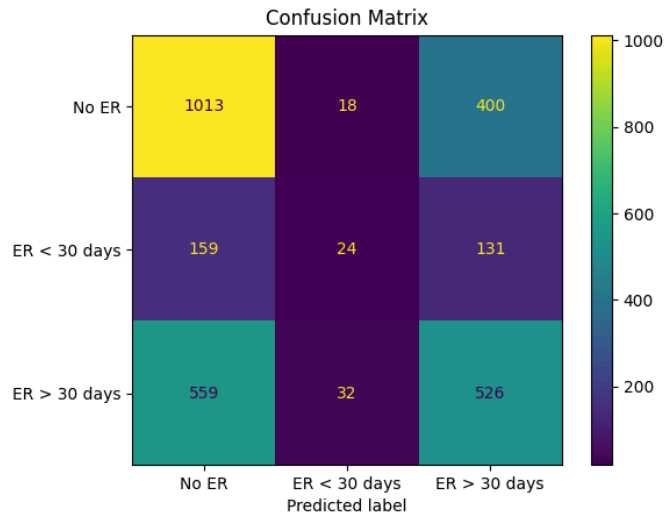
▼ plots

confusion_matrix.png

classification_report.txt

plots/confusion_matrix.png 24.93KB

Path: mlflow-artifacts:/0/59d67efeced349e189480c803e52cafa/artifacts/plots/confusion_matrix.png



Running ray_tune through the UI

This is the easier way to run ray_tune. In the UI, go here:



Train

Train the machine learning model with sample data derived from the Mediwatch dataset.

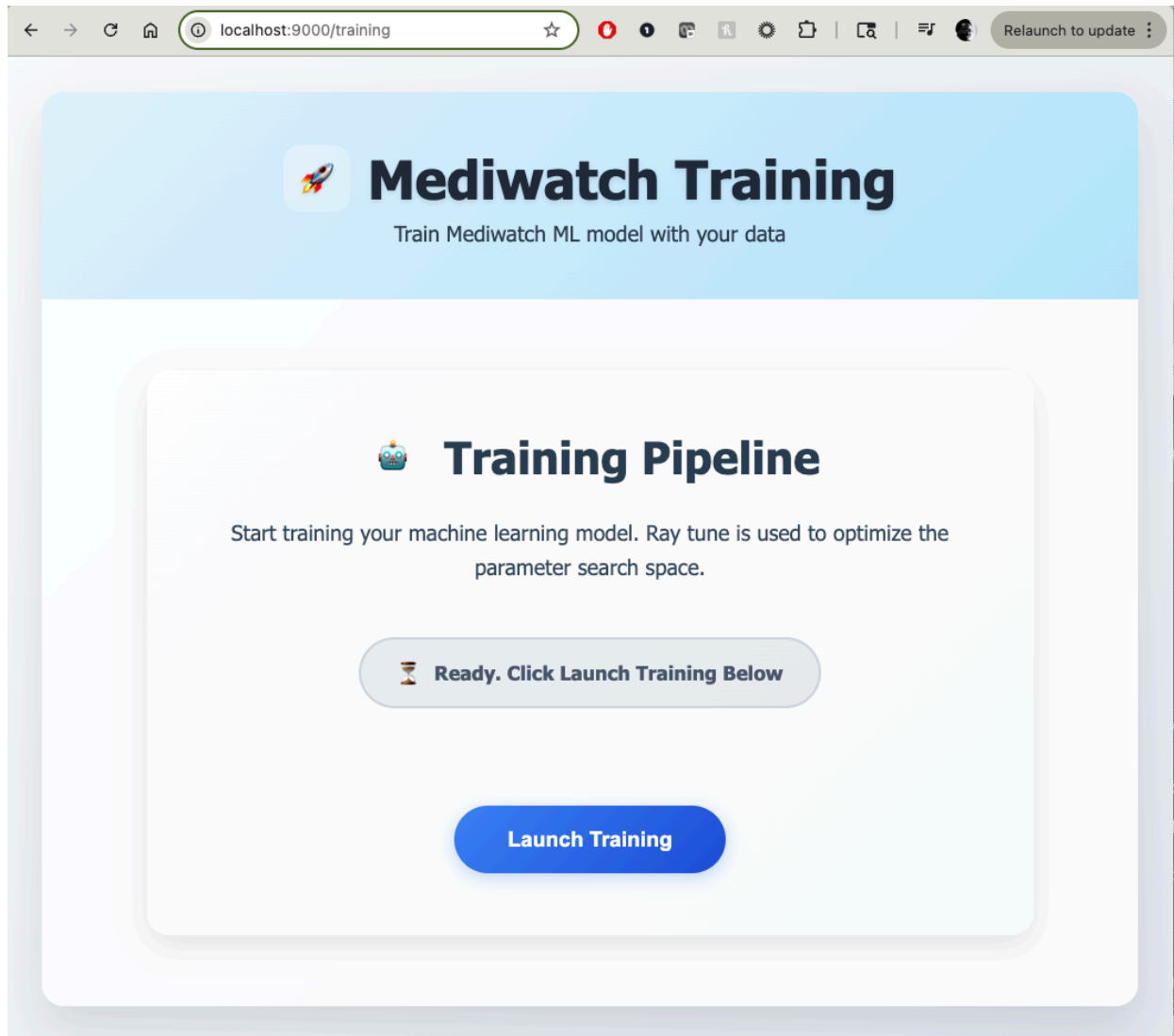
- ✓ Custom model training
- ✓ Progress monitoring with MLFlow
- ✓ Hyperparameter tuning with Ray-tune
- ✓ Performance metrics

Start Training

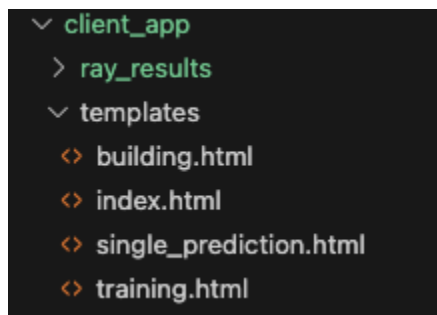
And click on the Start Training button.

Training endpoint

This takes you to the **/training** endpoint shown here



This page is rendered from the training.html template in `client_app/templates/training.html`



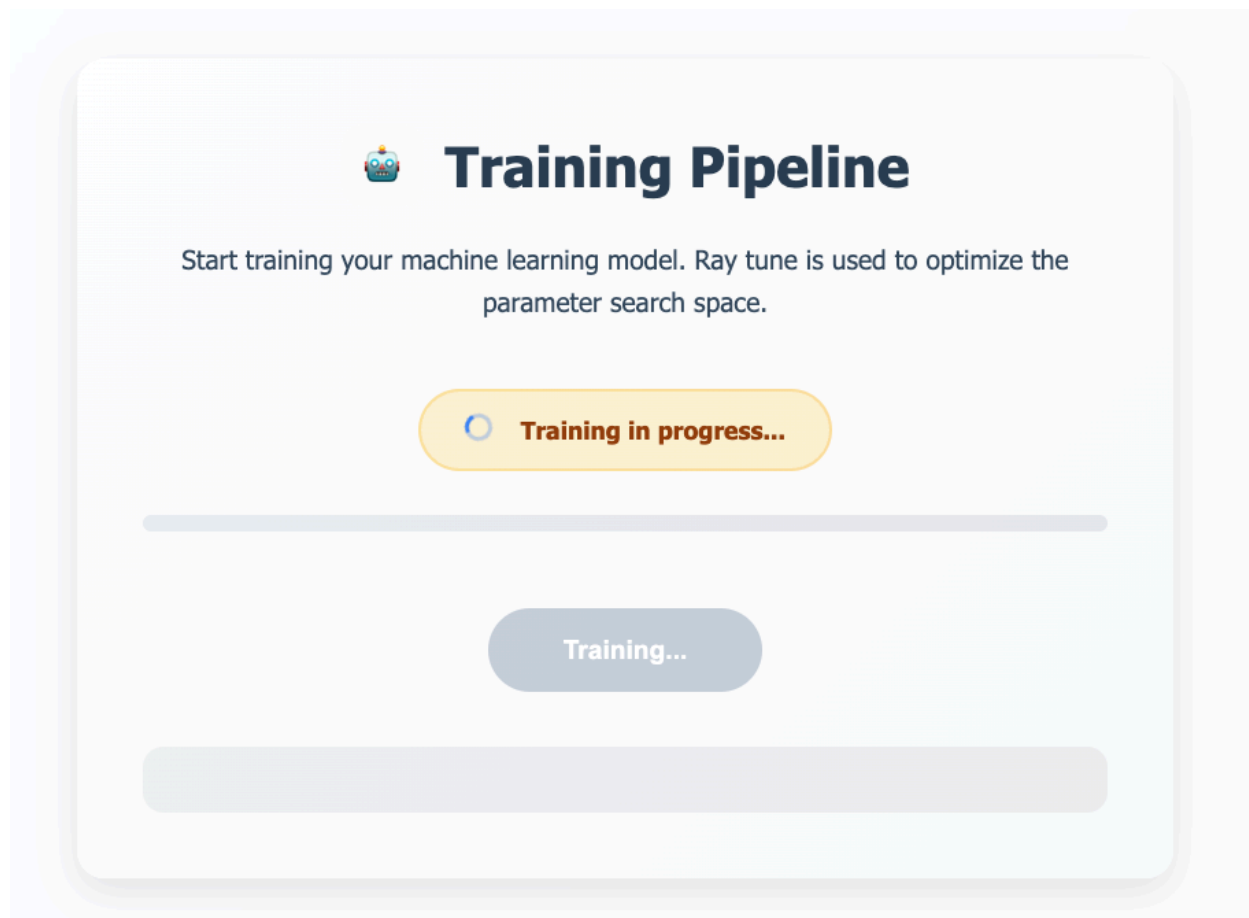
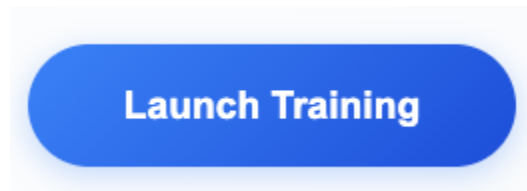
In Fast API, the `/training` renders the training page above.

GET	/training	Training
GET	/start_training	Start Training

The start_training runs ray_tune and returns logs IN the UI:

Start Training

Click on the Launch Training button



Logs are returned in line in the browser...

```
[9:12:35 PM] 2 0.50 0.48 0.49 1117
[9:12:35 PM]
[9:12:35 PM] accuracy 0.55 2862
[9:12:35 PM] macro avg 0.44 0.41 0.40 2862
[9:12:35 PM] weighted avg 0.52 0.55 0.52 2862
[9:12:35 PM]
[9:12:35 PM] 🏠 View run luminous-roo-813 at:
http://localhost:5000/#/experiments/0/runs/3dbecfaadef540fcb2332a01b1662eb4
[9:12:35 PM] ✍️ View experiment at: http://localhost:5000/#/experiments/0
[9:12:35 PM] Time taken for Ray tune (seconds): 37
[9:12:35 PM] Training completed
```

When completed, you get the button that takes you to mlflow

Training Complete

Start training your machine learning model. Ray tune is used to optimize the parameter search space.

✓ Training completed successfully

Launch Training

Track Training in MLflow

```
[9:12:35 PM] Training final model with best hyperparameters...
[9:12:35 PM]
[9:12:35 PM] Final Model Performance:
[9:12:35 PM] XGBoost Accuracy: 0.5506638714185884
[9:12:35 PM] XGBoost Class 1 Recall: 0.03821656050955414
[9:12:35 PM] precision recall f1-score support
[9:12:35 PM]
[9:12:35 PM] 0 0.59 0.72 0.65 1431
[9:12:35 PM] 1 0.21 0.04 0.06 314
[9:12:35 PM] 2 0.50 0.48 0.49 1117
[9:12:35 PM]
[9:12:35 PM] accuracy 0.55 2862
[9:12:35 PM] precision recall f1-score support
```

Where you can see your completed training run

2.0

Default >

luminous-roo-813

Overview Model metrics System metrics Traces Artifacts

Description 

No description

Details

Created at	10/05/2025, 09:11:53 PM
Created by	victor
Experiment ID	0 
Status	 Finished
Run ID	3dbecfaadef540fcb2332a01b1662eb4 
Duration	1.1s
Datasets used	—
Tags	Add tags
Source	 ray_tune.py  1d32d50
Logged models	—
Registered models	—
Registered prompts	—

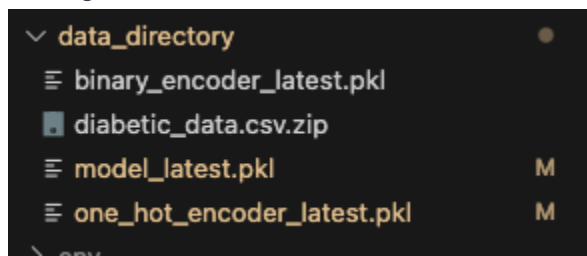
Metrics (2)

 Search metrics

Parameters (10)

 Search parameters

Note that the ray_tune training run will update the models in the model directory just like the basic training run did:



The start_training endpoint code is in the `client_app/ClientApp.py`. Here is a screenshot of part of that function:

```

# Start training, return logs to front end
@app.get("/start_training")
async def start_training():
    """Start training and return real-time logs via SSE"""

    async def generate_training_logs() -> AsyncGenerator[str, None]:
        # Your actual training command
        command = ["python3", clApp.ray_tune_script] # Replace with your training script
        process = subprocess.Popen(
            command,
            stdout=subprocess.PIPE,
            stderr=subprocess.STDOUT,
            universal_newlines=True,
            bufsize=1
        )

        try:
            while True:
                output = process.stdout.readline()
                if output == '' and process.poll() is not None:

```

Model Packaging and Deployment

FastAPI

Lets revisit FastAPI again in some more detail

Its available at port 9000 when **ClientApp.py** is running:

These are the endpoints

FastAPI

0.1.0 OAS 3.1

/openapi.json

default

GET	/	Home
GET	/health	Health
GET	/single_prediction	Single Prediction Page
POST	/single_prediction	Single Prediction Post
GET	/training	Training
GET	/start_training	Start Training
GET	/building	Building
POST	/dockerize	Dockerize
GET	/dockerize_stream	Dockerize Stream

And below are related schemas:

Schemas

DockerizeResponse ^ Collapse all object

```
status* string
message* string
image_name string
logs > Expand all array<string>
```

HTTPValidationError > Expand all object

MediwatchSubject ^ Collapse all object

```
race* string
gender* string
age* string
weight > Expand all (string | null)
admission_type_id* integer
discharge_disposition_id* integer
admission_source_id* integer
time_in_hospital* integer
payer_code > Expand all (string | null)
medical_specialty > Expand all (string | null)
num_lab_procedures* integer
num_procedures* integer
num_medications* integer
number_outpatient* integer
```

We can run a couple of endpoints to see what is available

Test root endpoint

Here is a sample run against the root '/' endpoint

GET / Home

Render homepage

Parameters

Cancel

No parameters

ExecuteClear

Responses

Curl

```
curl -X 'GET' \
'http://localhost:9000/' \
-H 'accept: text/html'
```

Request URL

```
http://localhost:9000/
```

Server response

CodeDetails

200

Response body

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>My Project</title>
  <style>
    * {
      margin: 0;
      padding: 0;
      box-sizing: border-box;
    }
  
```

Test Health Endpoint

Here is a sample run against the health endpoint

GET /health Health

Return OK if the ClientApp is up and running

Parameters

Cancel

No parameters

ExecuteClear

Responses

Curl

```
curl -X 'GET' \
'http://localhost:9000/health' \
-H 'accept: text/html'
```

Request URL

```
http://localhost:9000/health
```

Server response

CodeDetails

200

Response body

OK

Download

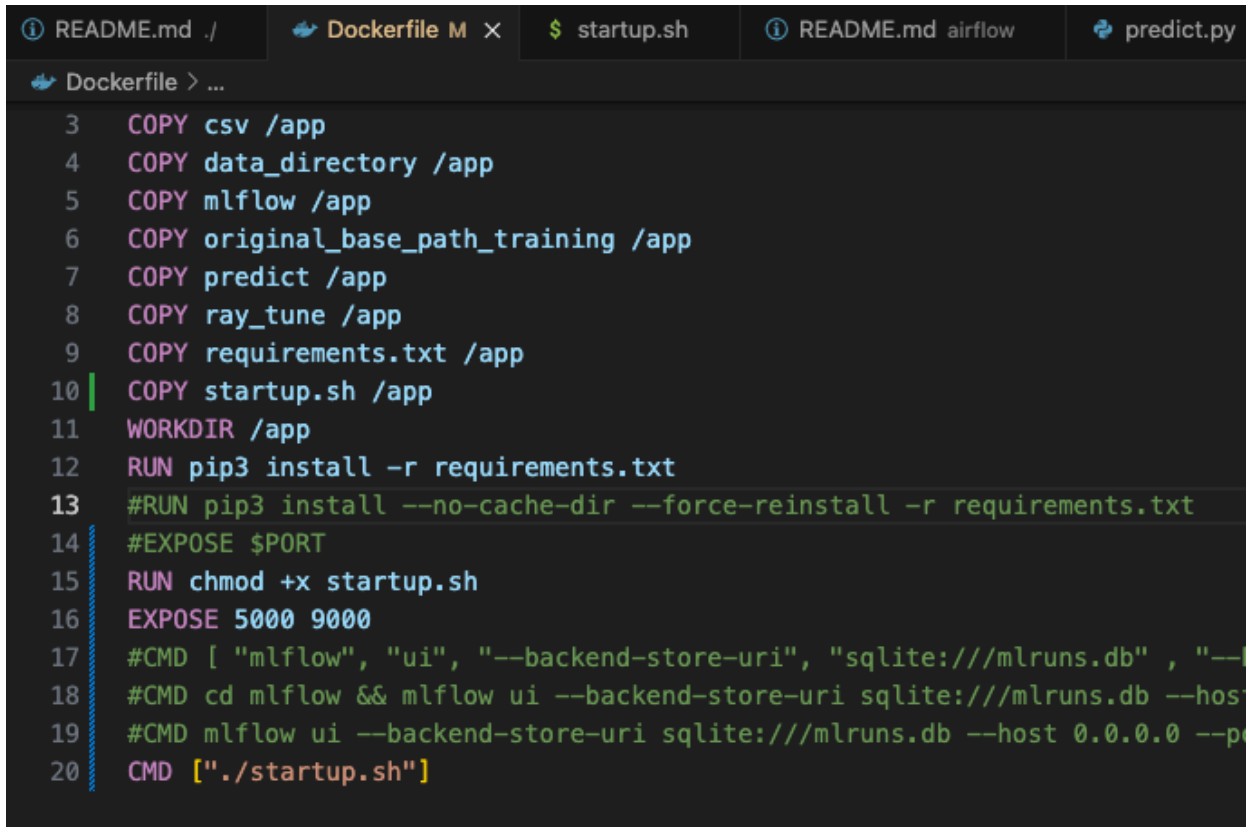
Response headers

```
content-length: 2
content-type: text/html; charset=utf-8
date: Mon, 06 Oct 2025 23:56:24 GMT
server: uvicorn
```

Generate Docker Image

Dockerfile

First, look at the Dockerfile



```
3 COPY csv /app
4 COPY data_directory /app
5 COPY mlflow /app
6 COPY original_base_path_training /app
7 COPY predict /app
8 COPY ray_tune /app
9 COPY requirements.txt /app
10 COPY startup.sh /app
11 WORKDIR /app
12 RUN pip3 install -r requirements.txt
13 #RUN pip3 install --no-cache-dir --force-reinstall -r requirements.txt
14 #EXPOSE $PORT
15 RUN chmod +x startup.sh
16 EXPOSE 5000 9000
17 #CMD [ "mlflow", "ui", "--backend-store-uri", "sqlite:///mlruns.db" , "--l
18 #CMD cd mlflow && mlflow ui --backend-store-uri sqlite:///mlruns.db --hos
19 #CMD mlflow ui --backend-store-uri sqlite:///mlruns.db --host 0.0.0.0 --p
20 CMD ["/startup.sh"]
```

I open ports 5000, 9000 for mlflow and the UI respectively.

Startup.sh

I coordinate startup of these scripts in the [‘startup.sh’](#) file as follows:

```
$ startup.sh
1  #!/bin/bash
2
3  # Start MLflow in the background
4  cd mlflow
5  mlflow ui --backend-store-uri sqlite:///mlruns.db --host 0.0.0.0 --port 5000 &
6
7  # Start the client app
8  cd ..
9  cd client_app && python3 ClientApp.py
```

Docker images before build

Look at the available docker images:

```
(base) victor@victor-Asus-PC:~/Documents/interview_kickstart/Capstone/Capstone-MLOPS-Mediwatch$ docker image ls
REPOSITORY          TAG                IMAGE ID           CREATED            SIZE
test_image1759979433 latest            bd6bda391ecc       24 minutes ago    11.9GB
redis                latest            9d1fe3a9a889       7 weeks ago       137MB
postgres            13                645e932c27f7       7 weeks ago       439MB
victor_iris_k8s      latest            e8755e76702b       3 months ago      5.73GB
victor_dog_cat       latest            9b35a1e0d610       3 months ago      3.59GB
gcr.io/k8s-minikube/kicbase v0.0.47          795ea6a69ce6       4 months ago      1.31GB
apache/airflow       2.7.2            dad28d0405d6       22 months ago     1.54GB
apache/airflow       2.7.1-python3.11 4fb802b917bc       22 months ago     1.51GB
```

Run Docker build from UI

Then go to the UI and click on the Build Section



Build

Build a Docker image containing the Mediwatch application.



Model Building

Start Building

Click on the Start Building Button

You get this page which allows you to start the Docker build



Mediwatch Build

Build Docker containers for your application



Build Pipeline

Start building your Docker container. The containerization process will package your application with all dependencies.



Ready. Click Launch Build Below

Launch Build

Click on the Launch build button



Build Pipeline

Start building your Docker container. The containerization process will package your application with all dependencies.



Build in progress...

Building...

(it will take a few mins to start logs...)



Build in progress...

Building...

```
[8:31:55 PM] #15 243.9 Downloading pytz-2025.2-py2.py3-none-any.whl (509
kB)
[8:31:55 PM] #15 243.9 Downloading pyyaml-6.0.3-cp312-cp312-
manylinux2014_x86_64.manylinux_2_17_x86_64.manylinux_2_28_x86_64.whl (807
kB)
[8:31:55 PM] #15 244.0 807.9/807.9
kB 12.2 MB/s eta 0:00:00
[8:31:55 PM] #15 244.0 Downloading requests-2.32.5-py3-none-any.whl (64 kB)
[8:31:55 PM] #15 244.0 Downloading rich-14.1.0-py3-none-any.whl (243 kB)
[8:31:55 PM] #15 244.1 Downloading scipy-1.16.2-cp312-cp312-
manylinux2014_x86_64.manylinux_2_17_x86_64.whl (35.7 MB)
[8:31:55 PM] #15 245.9 35.7/35.7
MB 19.4 MB/s eta 0:00:00
```

About ~ 10 minutes later

✓ **Build completed successfully**

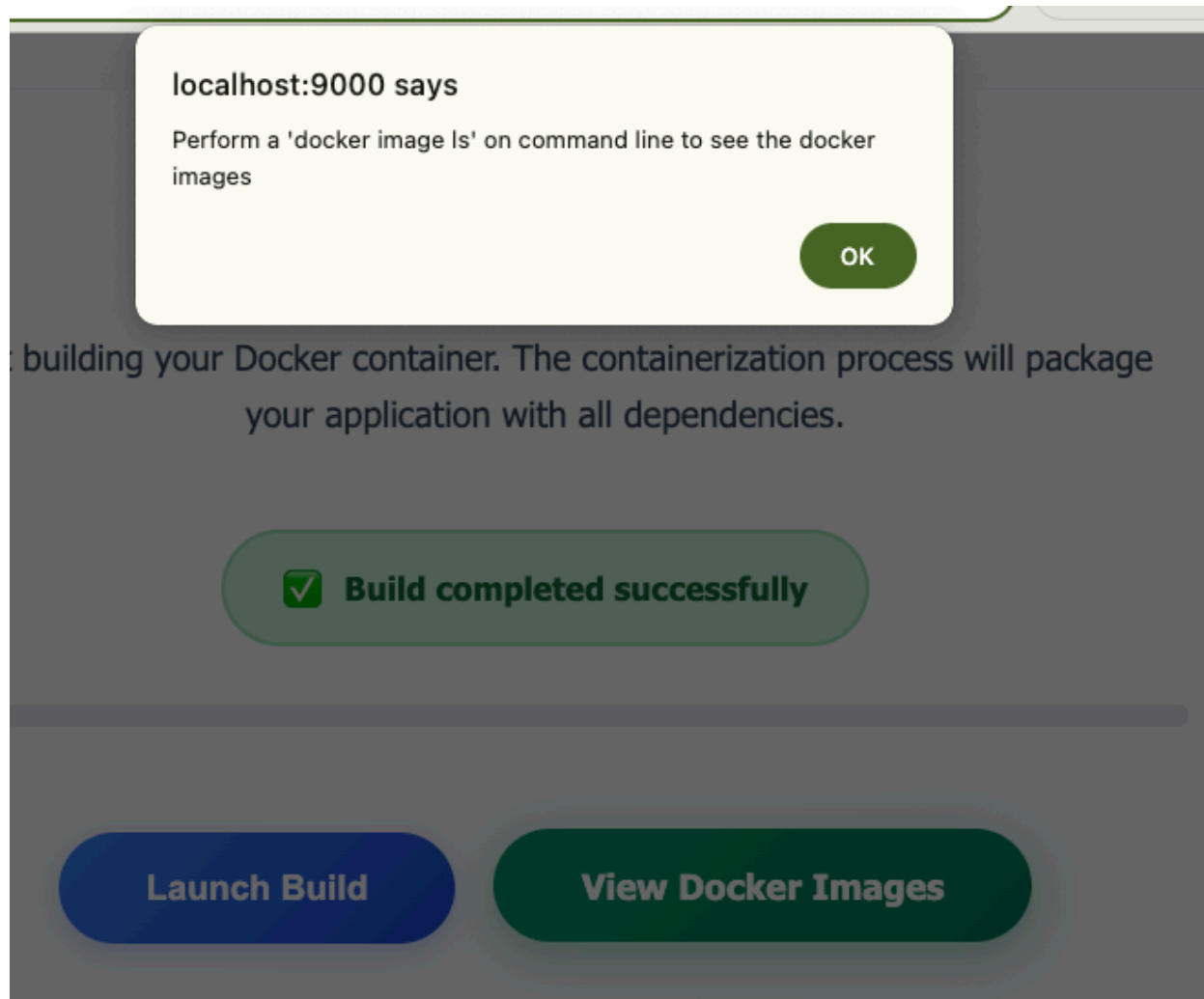
Launch Build

View Docker Images

```
[8:42:00 PM] #15 788.8 [notice] To update, run: pip install --upgrade pip
[8:42:00 PM] #15 DONE 795.3s
[8:42:00 PM]
[8:42:00 PM] #16 exporting to image
[8:42:00 PM] #16 exporting layers
[8:42:00 PM] #16 exporting layers 54.9s done
[8:42:00 PM] #16 writing image
sha256:ebfe25a462400146cc66ac8a9e0f77f22752d42749549b2fe2690b4082e59867
0.0s done
[8:42:00 PM] #16 naming to docker.io/library/test_image1759796824 0.1s done
[8:42:00 PM] #16 DONE 55.1s
[8:42:00 PM] Docker image test_image1759796824 built successfully!
```

Also ...

The 'View Docker image' gives this useful message:



And when you run the command:

Docker images after build

```
• (base) victor@victor-Asus-PC:~/Documents/interview_kickstart/Capstone/Capstone-MLOPS-Mediwatch$ docker image ls
REPOSITORY          TAG         IMAGE ID      CREATED       SIZE
test_image1759981921 latest      33e018f69e5f  About a minute ago  11.9GB
test_image1759979433 latest      bd6bda391ecc  42 minutes ago  11.9GB
redis                latest      9d1fe3a9a889  7 weeks ago    137MB
postgres             13         645e932c27f7  7 weeks ago    439MB
victor_iris_k8s      latest     e8755e76702b  3 months ago    5.73GB
victor_dog_cat       latest     9b35a1e0d610  3 months ago    3.59GB
gcr.io/k8s-minikube/kicbase v0.0.47    795ea6a69ce6  4 months ago    1.31GB
apache/airflow       2.7.2      dad28d0405d6  22 months ago    1.54GB
apache/airflow       2.7.1-python3.11 4fb802b917bc  22 months ago    1.51GB
○ (base) victor@victor-Asus-PC:~/Documents/interview_kickstart/Capstone/Capstone-MLOPS-Mediwatch$
```

The new docker image is available at the top of the list

Run Docker image

Lets run the Docker image now that it has been built...

For my image `test_image1759981921`

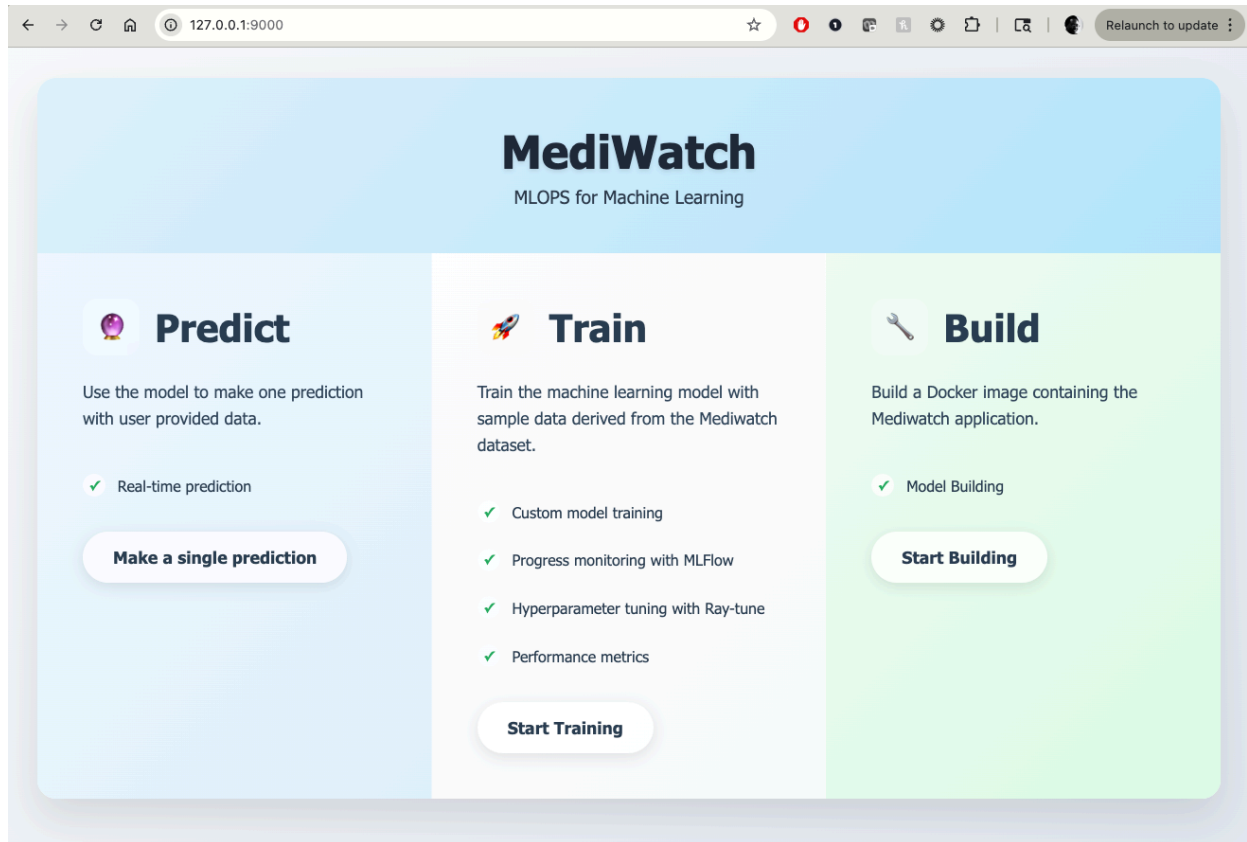
Lets run it ...

```
sudo docker run -p 5000:5000 -p 9000:9000 test_image1759981921
```

```
(base) victor@victor-Asus-PC:~/Documents/interview_kickstart/Capstone/Capstone-MLOPS-Mediwatch$
(base) victor@victor-Asus-PC:~/Documents/interview_kickstart/Capstone/Capstone-MLOPS-Mediwatch$ sudo docker run -p 5000:5000 -p 9000:9000 test_image1759981921
[sudo] password for victor:
INFO: Started server process [8]
INFO: Waiting for application startup.
INFO: Application startup complete.
INFO: Uvicorn running on http://0.0.0.0:9000 (Press CTRL+C to quit)
2025/10/09 04:06:53 INFO mlflow.store.db.utils: Creating initial MLflow database tables...
2025/10/09 04:06:53 INFO mlflow.store.db.utils: Updating database tables
INFO [alembic.runtime.migration] Context impl SQLiteImpl.
INFO [alembic.runtime.migration] Will assume non-transactional DDL.
INFO [alembic.runtime.migration] Context impl SQLiteImpl.
INFO [alembic.runtime.migration] Will assume non-transactional DDL.
2025/10/09 04:06:54 INFO mlflow.store.db.utils: Creating initial MLflow database tables...
2025/10/09 04:06:54 INFO mlflow.store.db.utils: Updating database tables
INFO [alembic.runtime.migration] Context impl SQLiteImpl.
INFO [alembic.runtime.migration] Will assume non-transactional DDL.
[2025-10-09 04:06:54 +0000] [62] [INFO] Starting gunicorn 23.0.0
[2025-10-09 04:06:54 +0000] [62] [INFO] Listening at: http://0.0.0.0:5000 (62)
[2025-10-09 04:06:54 +0000] [62] [INFO] Using worker: sync
[2025-10-09 04:06:54 +0000] [63] [INFO] Booting worker with pid: 63
[2025-10-09 04:06:54 +0000] [64] [INFO] Booting worker with pid: 64
[2025-10-09 04:06:54 +0000] [65] [INFO] Booting worker with pid: 65
[2025-10-09 04:06:54 +0000] [66] [INFO] Booting worker with pid: 66
Warning: ray_tune.py not found at /app/client_app../ray_tune/ray_tune.py
INFO: 172.17.0.1:48776 - "GET / HTTP/1.1" 200 OK
INFO: 172.17.0.1:48780 - "GET /single_prediction HTTP/1.1" 200 OK
/usr/local/lib/python3.12/site-packages/pydantic/_internal/_generate_schema.py:2249: UnsupportedFieldAttributeWarning: The 'alias' attribute with value 'glyburide-metformin' was provided to the 'Field()' function, which has no effect in the context it was used. 'alias' is field-specific metadata, and can only be attached to a model field using 'Annotated' metadata or by assignment. This may have happened because an 'Annotated' type alias using the 'type' statement was used, or if the 'Field()' function was attached to a single member of a union type.
  warnings.warn(
/usr/local/lib/python3.12/site-packages/pydantic/_internal/_generate_schema.py:2249: UnsupportedFieldAttributeWarning: The 'alias' attribute with value 'glipizide-metformin' was provided to the 'Field()' function, which has no effect in the context it was used. 'alias' is f
```

Predict on Docker image

NOTE: Please Don't run Build or Train on the docker image. (I didn't have time to modify the UI to only show the Predict section)



And then...



Predict

Use the model to make one prediction with user provided data.



Real-time prediction

Make a single prediction

And then...



Mediwatch Single Prediction Form

PATIENT INFORMATION

Race

Select Race



Gender

Select Gender



Age *

[10-20]



Weight

ADMISSION INFORMATION

Admission Type ID

0

Discharge Disposition ID

0

Admission Source ID

0

Time in Hospital (days)

0

Payer Code

None

Medical Specialty

The screenshot shows a web application interface with a light gray background. At the top, there is a section titled "ADDITIONAL INFORMATION" in bold. Below this title, there are two labels: "Change in Medication" and "Diabetes Medication Prescribed". Each label is followed by a white dropdown menu with a small downward arrow on the right. Both dropdown menus currently display the word "Select". Below the "ADDITIONAL INFORMATION" section, there is a horizontal line. Underneath the line, there is a blue button with rounded corners and a gradient, containing the text "MAKE SINGLE PREDICTION" in white. Below the button, there is a section titled "PREDICTION RESULTS" in bold. Underneath this title, there is a white rectangular box with a thin gray border containing the text "No return to ER".

Success!

Stop Docker container

Use

```
docker stop <container_id>
```

```
docker rm <container_id>
```

```

no changes added to commit (use "git add" and/or "git commit -a")
(base) victor@victor-Asus-PC:~/Documents/interview_kickstart/Capstone/Capstone-MLOPS-Mediawatch$ docker ps -a
CONTAINER ID   IMAGE                                COMMAND                  CREATED        STATUS        PORTS
b1777692ad0d   test_image1759981921               "./startup.sh"          9 minutes ago  Up 9 minutes  0.0.0.0:5000->5000/tcp, [::]:5000->5000/tcp, 0.0.0.0:5
]:9000->9000/tcp    musing_bassi
(base) victor@victor-Asus-PC:~/Documents/interview_kickstart/Capstone/Capstone-MLOPS-Mediawatch$ docker stop b1777692ad0d

b1777692ad0d
(base) victor@victor-Asus-PC:~/Documents/interview_kickstart/Capstone/Capstone-MLOPS-Mediawatch$
(base) victor@victor-Asus-PC:~/Documents/interview_kickstart/Capstone/Capstone-MLOPS-Mediawatch$
(base) victor@victor-Asus-PC:~/Documents/interview_kickstart/Capstone/Capstone-MLOPS-Mediawatch$ docker ps -a
CONTAINER ID   IMAGE                                COMMAND                  CREATED        STATUS        PORTS        NAMES
b1777692ad0d   test_image1759981921               "./startup.sh"          11 minutes ago  Exited (137) 3 seconds ago  musing_bassi
(base) victor@victor-Asus-PC:~/Documents/interview_kickstart/Capstone/Capstone-MLOPS-Mediawatch$ sudo docker stop b1777692ad0d
[sudo] password for victor:
b1777692ad0d
(base) victor@victor-Asus-PC:~/Documents/interview_kickstart/Capstone/Capstone-MLOPS-Mediawatch$
(base) victor@victor-Asus-PC:~/Documents/interview_kickstart/Capstone/Capstone-MLOPS-Mediawatch$ docker ps -a
CONTAINER ID   IMAGE                                COMMAND                  CREATED        STATUS        PORTS        NAMES
b1777692ad0d   test_image1759981921               "./startup.sh"          11 minutes ago  Exited (137) 19 seconds ago  musing_bassi
(base) victor@victor-Asus-PC:~/Documents/interview_kickstart/Capstone/Capstone-MLOPS-Mediawatch$
(base) victor@victor-Asus-PC:~/Documents/interview_kickstart/Capstone/Capstone-MLOPS-Mediawatch$ docker rm b1777692ad0d
b1777692ad0d
(base) victor@victor-Asus-PC:~/Documents/interview_kickstart/Capstone/Capstone-MLOPS-Mediawatch$
(base) victor@victor-Asus-PC:~/Documents/interview_kickstart/Capstone/Capstone-MLOPS-Mediawatch$
(base) victor@victor-Asus-PC:~/Documents/interview_kickstart/Capstone/Capstone-MLOPS-Mediawatch$ docker ps -a
CONTAINER ID   IMAGE                                COMMAND                  CREATED        STATUS        PORTS        NAMES
(base) victor@victor-Asus-PC:~/Documents/interview_kickstart/Capstone/Capstone-MLOPS-Mediawatch$

```

Kubernetes

Deploy the Docker image on a Kubernetes cluster for effective deployment.

The previous documentation shows building a docker image directly outside minikube

This image needs to be inside minikube in order to work with kubernetes

This can be tagged and copied into minikube as shown below:

And then tag

```
docker tag test_image1760066178:latest test_image1760066178:latest
```

Load this into minikube (takes a LONG while)

```
minikube image load test_image1760066178:latest
```

The problem here is that the copy is inefficient. I literally crashed my Desktop PC 2-3 times when the copy hung my entire OS.

The better way to do this is to build DIRECTLY in minikube. Here is how its done manually ...

Start MiniKube

minikube status

```
(base) victor@victor-Asus-PC:~/Documents/interview_kickstart/Capstone/Capstone-MLOPS-Mediawatch$ minikube status
minikube
type: Control Plane
host: Stopped
kubelet: Stopped
apiserver: Stopped
kubeconfig: Stopped
```

Here minikube is OFF, so we need to start it

minikube start

```
apache/airflow 2.7.1-python3.11 4fb8026917bc 22 months ago 1.51GB
(base) victor@victor-Asus-PC:~/Documents/interview_kickstart/Capstone/Capstone-MLOPS-Mediawatch$ minikube start
🐳 minikube v1.36.0 on Ubuntu 24.04
🔔 minikube 1.37.0 is available! Download it: https://github.com/kubernetes/minikube/releases/tag/v1.37.0
💡 To disable this notice, run: 'minikube config set WantUpdateNotification false'

🌟 Using the docker driver based on existing profile
👍 Starting "minikube" primary control-plane node in "minikube" cluster
📶 Pulling base image v0.0.47 ...
🔄 Restarting existing docker container for "minikube" ...
🔧 Preparing Kubernetes v1.33.1 on Docker 28.1.1 ...
🔍 Verifying Kubernetes components...
   ▪ Using image docker.io/kubernetesui/dashboard:v2.7.0
   ▪ Using image gcr.io/k8s-minikube/storage-provisioner:v5
   ▪ Using image docker.io/kubernetesui/metrics-scraper:v1.0.8
💡 Some dashboard features require the metrics-server addon. To enable all features please run:

    minikube addons enable metrics-server

🌟 Enabled addons: storage-provisioner, default-storageclass, dashboard
🏠 Done! kubectl is now configured to use "minikube" cluster and "default" namespace by default
```

And now ...

```
(base) victor@victor-Asus-PC:~/Documents/interview_kickstart/Capstone/Capstone-MLOPS-Mediawatch$ minikube status
minikube
type: Control Plane
host: Running
kubelet: Running
apiserver: Running
kubeconfig: Configured
```

Get into minikube shell and look at the images available

```
apache/airflow 2.7.1-python3.11 4fb8026917bc 22 months ago 1.51GB
(base) victor@victor-Asus-PC:~/Documents/interview_kickstart/Capstone/Capstone-MLOPS-Mediawatch/kube_deployment$ eval $(minikube docker-env)
(base) victor@victor-Asus-PC:~/Documents/interview_kickstart/Capstone/Capstone-MLOPS-Mediawatch/kube_deployment$ docker images
REPOSITORY          TAG          IMAGE ID       CREATED        SIZE
registry.k8s.io/kube-scheduler v1.33.1      398c985c0d95   5 months ago  73.4MB
registry.k8s.io/kube-controller-manager v1.33.1     eef43894fa110 5 months ago  94.6MB
registry.k8s.io/kube-apiserver v1.33.1     c6ab243b29f8 5 months ago  102MB
registry.k8s.io/kube-proxy v1.33.1     b79c189b052c 5 months ago  97.9MB
registry.k8s.io/etcd 3.5.21-0    499038711c08 6 months ago  153MB
registry.k8s.io/coredns/coredns v1.12.0     1cf5f116067c 10 months ago 70.1MB
registry.k8s.io/pause 3.10         873ed7510279 16 months ago 736kB
kubernetesui/dashboard <none>        07655ddf2eeb 3 years ago  246MB
kubernetesui/metrics-scraper <none>       115053965e86 3 years ago  43.8MB
gcr.io/k8s-minikube/storage-provisioner v5          6e38f40d628d 4 years ago  31.5MB
```


Build in minikube

This is using the kube_deployment/Deployment.yaml

```
docker build -t test_image_1000:latest .
```

```

(base) victor@victor-Asus-PC:~/Documents/interview_kickstart/Capstone/Capstone-MLOPS-Medivatch$ docker build -t test_image_1000:latest .
[+] Building 597.3s (19/19) FINISHED
=> [internal] load build definition from Dockerfile
=> transferring Dockerfile: 1.03kB
=> [internal] load metadata for docker.io/library/python:3.12.10
=> [internal] load .dockerignore
=> transferring context: 2B
=> [ 1/14] FROM docker.io/library/python:3.12.10@sha256:45d13f5cba9b0f9bdfaed658d3e4fcfb1694ca9a6d70ec07c9f3980a7dc8b26
=> resolve docker.io/library/python:3.12.10@sha256:45d13f5cba9b0f9bdfaed658d3e4fcfb1694ca9a6d70ec07c9f3980a7dc8b26
=> sha256:45d13f5cba9b0f9bdfaed658d3e4fcfb1694ca9a6d70ec07c9f3980a7dc8b26 9.73kB / 9.73kB
=> sha256:1b0cfae64d89d02be73031f3512c3504525ee74bbbf7671b1cdded24e7db5b5b9 6.46kB / 6.46kB
=> sha256:3ee6b9da19511e19f12262a4e8a59ad1d1a10ca7b8210badcf0605a200294964 48.49MB / 48.49MB
=> sha256:81951572f7959dbdc7345744de9d396c56a6ac95c1dae2451132fe1ba633ae3 2.33kB / 2.33kB
=> sha256:3792f7e9901b1b2608b72796c6881bf645480268eca4ac9a3b9219e050bb4d84 24.02MB / 24.02MB
=> sha256:79b2474da444365c81a95ede249f1d76310efdbec159729638737780c 64.40MB / 64.40MB
=> extracting sha256:3ee6b9da19511e19f12262a4e8a59ad1d1a10ca7b8210badcf0605a200294964
=> sha256:1b0cfae64d89d02be73031f3512c3504525ee74bbbf7671b1cdded24e7db5b5b9 6.46kB / 6.46kB
=> sha256:d2717412051f07b289dae4776d1d34f119b6c3743c390996341c1b3c15d417 6.16MB / 6.16MB
=> sha256:81951572f7959dbdc7345744de9d396c56a6ac95c1dae2451132fe1ba633ae3 2.33kB / 2.33kB
=> sha256:1290187bfbf4d7435a337f7a034ccec4ba186242e9f7599bae884bf332030928 249B / 249B
=> extracting sha256:3792f7e9901b1b2608b72796c6881bf645480268eca4ac9a3b9219e050bb4d84

```

...

...

```
330 if output:
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS
[+] Building 214.5s (16/18)
=> extracting sha256:1290187bfbf4dc7435a33f7a03f4cec4ba186242e9f7599bae884bb332030928
=> [internal] load build context
=> transferring context: 152.63MB
=> [2/14] COPY client_app /app/client_app
=> [3/14] COPY csv /app/csv
=> [4/14] COPY data_directory /app/data_directory
=> [5/14] COPY mflow /app/mflow
=> [6/14] COPY original_base_path_training /app/original_base_path_training
=> [7/14] COPY predict /app/predict
=> [8/14] COPY ray_tune /app/predict
=> [9/14] COPY requirements.txt /app/
=> [10/14] COPY startup.sh /app/
=> [11/14] WORKDIR /app
=> [12/14] RUN pip3 install --no-cache-dir torch==2.7.1 --index-url https://download.pytorch.org/whl/cpu
=> [13/14] RUN pip3 install --no-cache-dir -r requirements.txt && find /usr/local/lib/python3.12/site-packages -type d -name "tests" -exec rm -rf {} +
=> # Downloading rsa-4.1-py3-none-any.whl (34 kB)
=> # Downloading smmap-5.0.2-py3-none-any.whl (24 kB)
=> # Downloading tzdata-2025.2-py2-py3-none-any.whl (347 kB)
=> # Downloading pyasn1-0.6.1-py3-none-any.whl (83 kB)
=> # Building wheels for collected packages: pandas
=> # Building wheel for pandas (pyproject.toml): started
```

...

...

Completed:

```
22 EXPOSE 5000 9000
23 CMD [". /startup.sh"]

PROBLEMS 0 OUTPUT DEBUG CONSOLE TERMINAL PORTS 6
=> => extracting sha256:d2717412051f07b2b9dae4e776d1d34f19b63c47c39d30996341c1bc31a5d417 0.1s
=> => extracting sha256:e1817d4c9557090860f306ad5a2c5ef09d4e3cee68892dcff52529ac7ac38 0.1s
=> => extracting sha256:1290187bfbf4dc7435a33f7a03f4cec4ba186242e9f7599bae8846b332030928 0.0s
=> [internal] load build context 2.4s
=> => transferring context: 152.63MB 2.4s
=> [ 2/14] COPY client_app /app/client_app 1.1s
=> [ 3/14] COPY csv /app/csv 0.1s
=> [ 4/14] COPY data_directory /app/data_directory 0.0s
=> [ 5/14] COPY mlflow /app/mlflow 0.0s
=> [ 6/14] COPY original_base_path_training /app/original_base_path_training 0.1s
=> [ 7/14] COPY predict /app/predict 0.0s
=> [ 8/14] COPY ray_tune /app/predict 0.7s
=> [ 9/14] COPY requirements.txt /app/ 0.0s
=> [10/14] COPY startup.sh /app/ 0.0s
=> [11/14] WORKDIR app 0.0s
=> [12/14] RUN pip3 install --no-cache-dir torch==2.7.1 --index-url https://download.pytorch.org/whl/cpu 22.3s
=> [13/14] RUN pip3 install --no-cache-dir -r requirements.txt && find /usr/local/lib/python3.12/site-packages -type d -name "tests" -exec rm -rf {} + 550.0s
=> [14/14] RUN chmod +x startup.sh 0.1s
=> exporting to image 8.2s
=> exporting layers 8.7s
=> writing image sha256:485cfd0b87d4ec2ddbdb48147c8e56a427cbf43818b766ee7552a193e6b4be3 0.0s
=> naming to docker.io/library/test_image_1000:latest 0.0s
o (base) victor@victor-Asus-PC:~/Documents/interview_kickstart/Capstone/Capstone-MLOPS-Mediawatch$
```

Show built Image in minikube

```
(base) victor@victor-Asus-PC:~/Documents/interview_kickstart/Capstone/Capstone-MLOPS-Mediawatch$ docker images
REPOSITORY          TAG          IMAGE ID       CREATED        SIZE
test_image_1000     latest      485cfd0b87d4   2 minutes ago  3.77GB
registry.k8s.io/kube-apiserver    v1.33.1     c6ab243b29f8   5 months ago  102MB
registry.k8s.io/kube-controller-manager v1.33.1     ef43894fa110   5 months ago  94.6MB
registry.k8s.io/kube-scheduler    v1.33.1     398c985c0d95   5 months ago  73.4MB
registry.k8s.io/kube-proxy        v1.33.1     b79c189b052c   5 months ago  97.9MB
registry.k8s.io/etcd              3.5.21-0    499038711c08   6 months ago  153MB
registry.k8s.io/coredns/coredns   v1.12.0     1cf5f116067c   10 months ago 70.1MB
registry.k8s.io/pause             3.10        873ed7510279   16 months ago 736kB
kubernetesui/dashboard            <none>       07655ddf2eeb   3 years ago   246MB
kubernetesui/metrics-scraper      <none>       115053965e86   3 years ago   43.8MB
gcr.io/k8s-minikube/storage-provisioner v5          6e38f40d628d   4 years ago   31.5MB
(base) victor@victor-Asus-PC:~/Documents/interview_kickstart/Capstone/Capstone-MLOPS-Mediawatch$
```

Update the Deployment.yaml

With the new image

```
test_image_1000:latest
```

```

kube_deployment > ! Deployment.yaml
3  metadata:
4
5      labels:
6          app: victor-mediwatch
7  spec:
8      replicas: 1
9      selector:
10         matchLabels:
11             app: victor-mediwatch
12     template:
13         metadata:
14             labels:
15                 app: victor-mediwatch
16         spec:
17             containers:
18                 - name: victor-mediwatch-container
19                   #image: test_image1760212598:latest
20                   image: test_image_1000:latest
21                   imagePullPolicy: Never
22             ports:
23                 - containerPort: 9000
24                 - containerPort: 5000
25             restartPolicy: Always
26

```

Apply the deployment

```
kubectl apply -f Deployment.yaml
```

```

-rw-rw-r-- 1 victor victor 5235 Oct 11 11:58 test_kubernetes.py
• (base) victor@victor-Asus-PC:~/Documents/interview_kickstart/Capstone/Capstone-MLOPS-Mediwatch/kube_deployment$ kubectl apply -f Deployment.yaml
deployment.apps/victor-mediwatch-deployment configured
• (base) victor@victor-Asus-PC:~/Documents/interview_kickstart/Capstone/Capstone-MLOPS-Mediwatch/kube_deployment$ kubectl get pods
NAME                                READY   STATUS    RESTARTS   AGE
victor-mediwatch-deployment-6985675756-dv28k  1/1     Running   0           6s

```

Apply the Service..

```
victor-mediwatch-deployment-6985675756-dv28k 1/1 Running 0 6s
• (base) victor@victor-Asus-PC:~/Documents/interview_kickstart/Capstone/Capstone-MLOPS-Mediwatch/kube_deployment$ kubectl apply -f Service.yaml
service/victor-mediwatch-service unchanged
• (base) victor@victor-Asus-PC:~/Documents/interview_kickstart/Capstone/Capstone-MLOPS-Mediwatch/kube_deployment$ kubectl get pods
NAME                                READY STATUS RESTARTS AGE
victor-mediwatch-deployment-6985675756-dv28k 1/1 Running 0 58s
○ (base) victor@victor-Asus-PC:~/Documents/interview_kickstart/Capstone/Capstone-MLOPS-Mediwatch/kube_deployment$
```

Check services and ports...

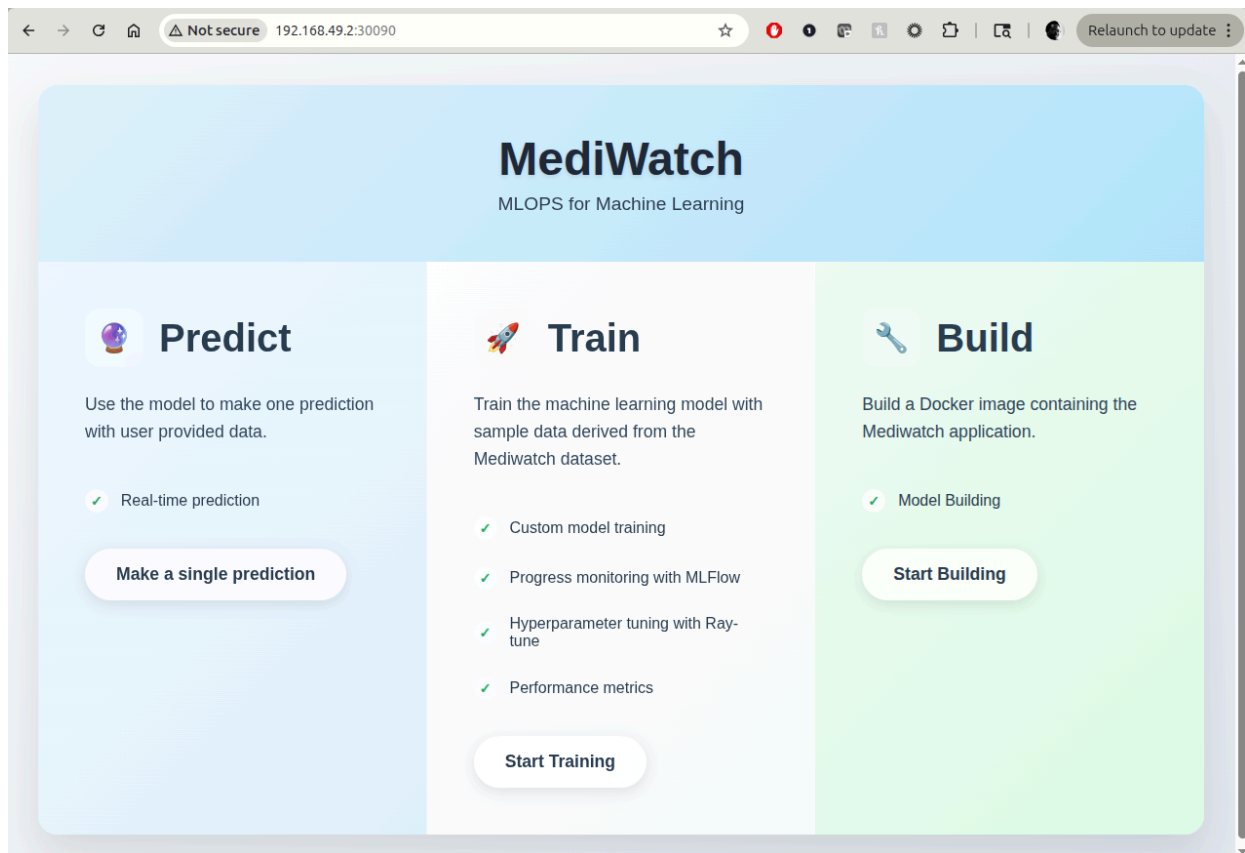
```
kubectl get nodes -o wide
```

```
victor-mediwatch-deployment-6985675756-dv28k 1/1 Running 0 58s
• (base) victor@victor-Asus-PC:~/Documents/interview_kickstart/Capstone/Capstone-MLOPS-Mediwatch/kube_deployment$ kubectl get nodes -o wide
NAME STATUS ROLES AGE VERSION INTERNAL-IP EXTERNAL-IP OS-IMAGE KERNEL-VERSION CONTAINER-RUNTIME
minikube Ready control-plane 3d13h v1.33.1 192.168.49.2 <none> Ubuntu 22.04.5 LTS 6.8.0-85-generic docker://28.1.1
○ (base) victor@victor-Asus-PC:~/Documents/interview_kickstart/Capstone/Capstone-MLOPS-Mediwatch/kube_deployment$
```

Kubernetes services

NOTE: Please Don't run Build or Train on the docker image. (I didn't have time to modify the UI to only show the Predict section)

UI in kubernetes at 30090





Mediwatch Single Prediction Form

PATIENT INFORMATION

Race

African American

Gender

Select Gender

Age *

[20-30]

Weight

ADMISSION INFORMATION

Admission Type ID

0

Discharge Disposition ID

0

Admission Source ID

0

Time in Hospital (days)

0

Payer Code

None

Medical Specialty

Citoglipton *
No

Insulin *
No

COMBINATION MEDICATIONS

Glyburide-Metformin *
No

Glipizide-Metformin *
No

Glimepiride-Pioglitazone *
No

Metformin-Rosiglitazone *
No

Metformin-Pioglitazone *
No

ADDITIONAL INFORMATION

Change in Medication
Select

Diabetes Medication Prescribed
Select

MAKE SINGLE PREDICTION

PREDICTION RESULTS

No return to ER

FastAPI 0.1.0 OAS 3.1

/openapi.json

default

GET	/	Home
GET	/health	Health
GET	/single_prediction	Single Prediction Page
POST	/single_prediction	Single Prediction Post
GET	/training	Training
GET	/start_training	Start Training
GET	/building	Building
POST	/dockerize	Dockerize
GET	/dockerize_stream	Dockerize Stream
POST	/generate-deployment	Generate Deployment
POST	/kubernetes_deploy	Kubernetes Deploy

Schemas

DockerizeResponse > Expand all object

HTTPValidationError > Expand all object

Note: training and build UI do not work in kubernetes!!!

The screenshot shows the MLflow 3.2.0 web interface. The left sidebar contains navigation links for 'New', 'Experiments', 'Models', and 'Prompts'. The main content area displays the details for an experiment named 'judicious-zebra-601'.

Experiment Details:

- Description:** No description
- Details Table:**

Created at	10/13/2025, 02:36:54 PM
Created by	victor
Experiment ID	0
Status	Finished
Run ID	8d665ff46f05450db4694c8f801730d8
Duration	0.7s
Datasets used	—
Tags	Add tags
Source	ray_tune.py -> b30bc8e
Logged models	—
Registered models	—
Registered prompts	—

Metrics (2):

Metric	Value
Train Accuracy	0.573025856044724
Train Class 1 Recall	0.03184713375796178

Parameters (10):

Parameter	Value
n_estimators	100
max_depth	5
learning_rate	0.17171
subsample	0.6608
colsample_bytree	0.8175

Turn off Kubernetes deployment

And when you are done:

... (set replicas to 0)

```
(base) victor@victor-Asus-PC:~/Documents/interview_kickstart/Capstone/Capstone-MLOPS-Mediwatch/kube_deployments$ kubectl scale --replicas=0 deployment/victor-mediwatch-deployment
deployment.apps/victor-mediwatch-deployment scaled
(base) victor@victor-Asus-PC:~/Documents/interview_kickstart/Capstone/Capstone-MLOPS-Mediwatch/kube_deployments$ kubectl get pods
NAME                                READY   STATUS    RESTARTS   AGE
victor-mediwatch-deployment-6985675756-dv28k  1/1     Terminating   0          13m
(base) victor@victor-Asus-PC:~/Documents/interview_kickstart/Capstone/Capstone-MLOPS-Mediwatch/kube_deployments$
```

Model Monitoring with Evidently

Data Drift Considerations

For this project, we are handed a static data file to use. I saved this locally as 'diabetic_data.csv'. For a refresher, let us take a look at the csv files directory:



The 'diabetic_data.csv' is divided into 6 files: chunk_1.csv ... chunk_6.csv. I ONLY train with chunk_1.csv and use 'chunk_6.csv' as my new data.

This has 2 implications

- 1) The model is probably undertrained because we gave it only 1/6 of the potential data it could use
- 2) Evidently is very likely to find drift when checking model performance between these 2 different sets of data
- 3) We can then very easily kick off the training_and_docker_build dag to rebuild and update the running instance in kubernetes.

Evidently

As mentioned, I am using evidently to detect drift.

Evidently requires Airflow to be up and running. It also needs the client_app to be running

Config File

Lets take a quick look at the evidently config file located at **evidently/config**:

```

evidently > config.py > ...
1 PROJECT_NAME = "Mediwatch Dashboard"
2
3 # Evidently Workspace path
4 WORKSPACE_PATH = "/home/victor/Documents/interview_kickstart/Capstone/Capstone-MLOPS-Mediwatch/evidently/evidently-data"
5
6 # Training data
7 TRAINING_DATA_FILE = "../csv/files/chunk_1.csv"
8
9 # 'New' data
10 NEW_DATA_FILE= "../csv/files/chunk_5.csv"
11
12 # Define threshold for ValueDrift (Jensen-Shannon distance)
13 # 0.2 is a Common threshold for Jensen-Shannon distance
14 JS_THRESHOLD=0.1
15
16 AIRFLOW_API_URL = 'http://192.168.50.6:8080/api/v1/dags/training_and_docker_build/dagRuns'
17 AIRFLOW_USERNAME = 'admin'
18 AIRFLOW_PASSWORD = 'admin123'
19

```

As mentioned previously, the training file is **chunk_1.csv** and the new data file is **chunk_6.csv**.

Also note the **WORKSPACE_PATH** and **AIRFLOW_API_URL** which need to be set up first.

Evidently Workspace

The evidently workspace is automatically set up in **evidently/evidently-data** . You just need to provide the full path as shown in the config file.

Check for drift

The script **evidently/infer_and_generate_tests_reports.py** will

- 1) Create and initialize the evidently-data directory if it does not exist and
- 2) Calculate the drift between the training data and the new data.
- 3) If it finds drift, it will trigger the **training_and_docker_build** which is described in detail in the Airflow section

This is how to run the script:

First, activate the env

```
conda activate ./env/
```

Then cd to the folder

```
cd evidently/
```

Then run the script

```
python3 infer_and_generate_tests_reports.py
```

Look at the output

```

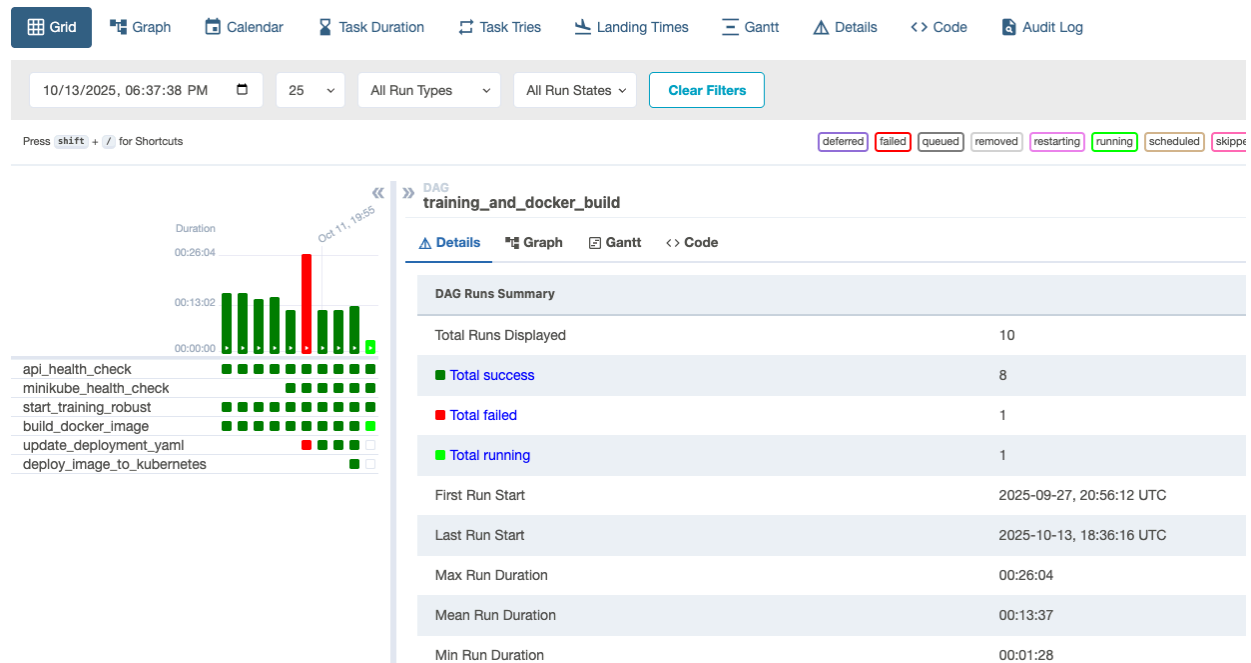
(/home/victor/Documents/interview_kickstart/Capstone/Capstone-MLOPS-Mediwatch/env) victor@victor-Asus-PC:~/Documents/interview_kickstart/Capstone/Capstone-MLOPS-Mediwatch$ S-Mediwatch/evidently$ python3 infer_and_generate_tests_reports.py
/home/victor/Documents/interview_kickstart/Capstone/Capstone-MLOPS-Mediwatch/env/lib/python3.10/site-packages/sklearn/preprocessing/_encoders.py:246: UserWarning: Found unknown categories in columns [4, 13, 14, 20, 21] during transform. These unknown categories will be encoded as all zeros
  warnings.warn(
Project: Mediwatch Dashboard already exists. Skip Creation.
Adding Report to project Mediwatch Dashboard

Drift Analysis Results:
Metric ValueDrift(column=readmitted_predictions) has drift and has a value of 0.15238455447137636
WARNING: Data drift detected in the dataset!
DAG has been triggered successfully.
(/home/victor/Documents/interview_kickstart/Capstone/Capstone-MLOPS-Mediwatch/env) victor@victor-Asus-PC:~/Documents/interview_kickstart/Capstone/Capstone-MLOPS-Mediwatch$ S-Mediwatch/evidently$

```

You can see the DAG was triggered

Also, look in airflow, you can see the DAG currently running



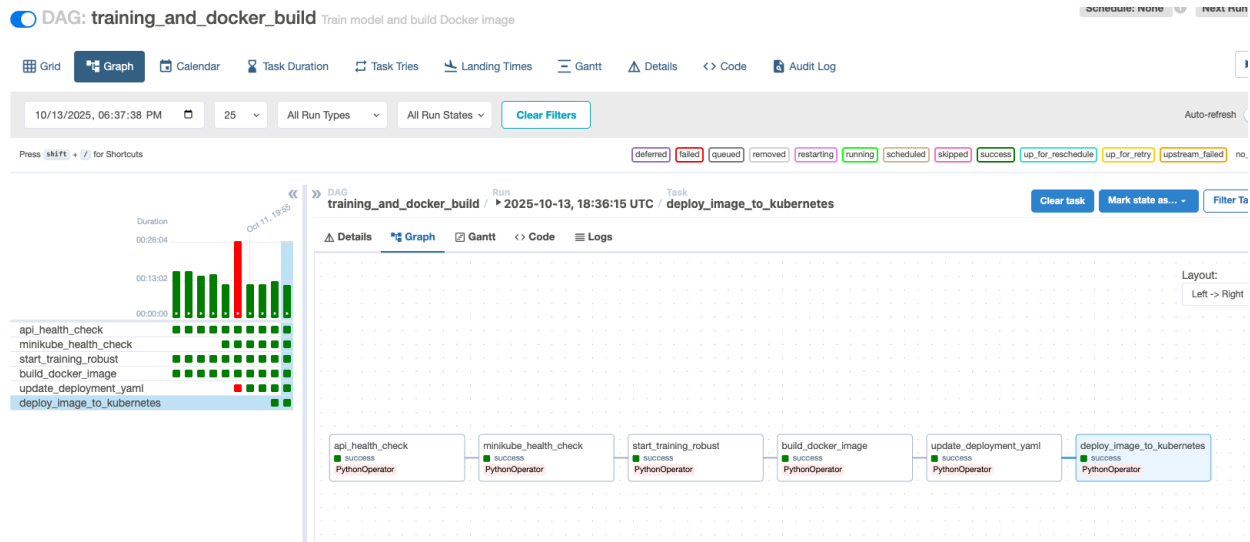
For reference, here are my docker images in minikube before the run:

```

(base) victor@victor-Asus-PC:~/Documents/interview_kickstart/Capstone/Capstone-MLOPS-Mediwatch$ docker images
REPOSITORY          TAG                 IMAGE ID            CREATED             SIZE
test_image1760378329 latest             e0df79810aab       27 minutes ago    3.78GB
test_image1760372126 latest             a722e9def9b6       2 hours ago       3.77GB
test_image1760369737 latest             1df71c62037b       3 hours ago       3.77GB
test_image_1000      latest             485cfd0b87d4       4 hours ago       3.77GB
registry.k8s.io/kube-controller-manager v1.33.1            ef43894fa110       5 months ago      94.6MB
registry.k8s.io/kube-scheduler v1.33.1            398c985c0d95       5 months ago      73.4MB
registry.k8s.io/kube-apiserver v1.33.1            c6ab243b29f8       5 months ago      102MB
registry.k8s.io/kube-proxy v1.33.1            b79c189b052c       5 months ago      97.9MB
registry.k8s.io/etcd 3.5.21-0           499038711c08       6 months ago      153MB
registry.k8s.io/coredns/coredns v1.12.0           1cf5f116067c       10 months ago     70.1MB
registry.k8s.io/pause 3.10               873ed7510279       16 months ago     736kB
kubernetesui/dashboard <none>             07655ddf2eeb       3 years ago        246MB
kubernetesui/metrics-scraper <none>            115053965e86       3 years ago        43.8MB
gcr.io/k8s-minikube/storage-provisioner v5               6e38f40d628d       4 years ago        31.5MB

```

And after the run completes:



with the new docker image in minikube:

```
(base) victor@victor-Asus-PC:~/Documents/interview_kickstart/Capstone/Capstone-MLOPS-Mediawatch$ docker images
REPOSITORY                                TAG      IMAGE ID      CREATED        SIZE
test_image1760380618                      latest   abf284b72488  6 minutes ago  3.77GB
test_image1760378329                      latest   e0df79810aab  44 minutes ago 3.78GB
test_image1760372126                      latest   a722e9def9b6  2 hours ago    3.77GB
test_image1760369737                      latest   1df71c62037b  3 hours ago    3.77GB
test_image_1000                          latest   485cf0b87d4   4 hours ago    3.77GB
registry.k8s.io/kube-apiserver            v1.33.1  c6ab243b29f8  5 months ago   102MB
registry.k8s.io/kube-controller-manager  v1.33.1  ef43894fa110  5 months ago   94.6MB
registry.k8s.io/kube-scheduler            v1.33.1  398c985c0d95  5 months ago   73.4MB
registry.k8s.io/kube-proxy                v1.33.1  b79c189b052c  5 months ago   97.9MB
registry.k8s.io/etcd                      3.5.21-0 499038711c08  6 months ago   153MB
registry.k8s.io/coredns/coredns           v1.12.0  1cf5f116067c  10 months ago  70.1MB
registry.k8s.io/pause                     3.10     873ed7510279  16 months ago  736kB
kubernetesui/dashboard                   <none>    07655ddf2eeb  3 years ago    246MB
kubernetesui/metrics-scraper              <none>    115053965e86  3 years ago    43.8MB
gcr.io/k8s-minikube/storage-provisioner   v5       6e38f40d628d  4 years ago    31.5MB
(base) victor@victor-Asus-PC:~/Documents/interview_kickstart/Capstone/Capstone-MLOPS-Mediawatch$
```

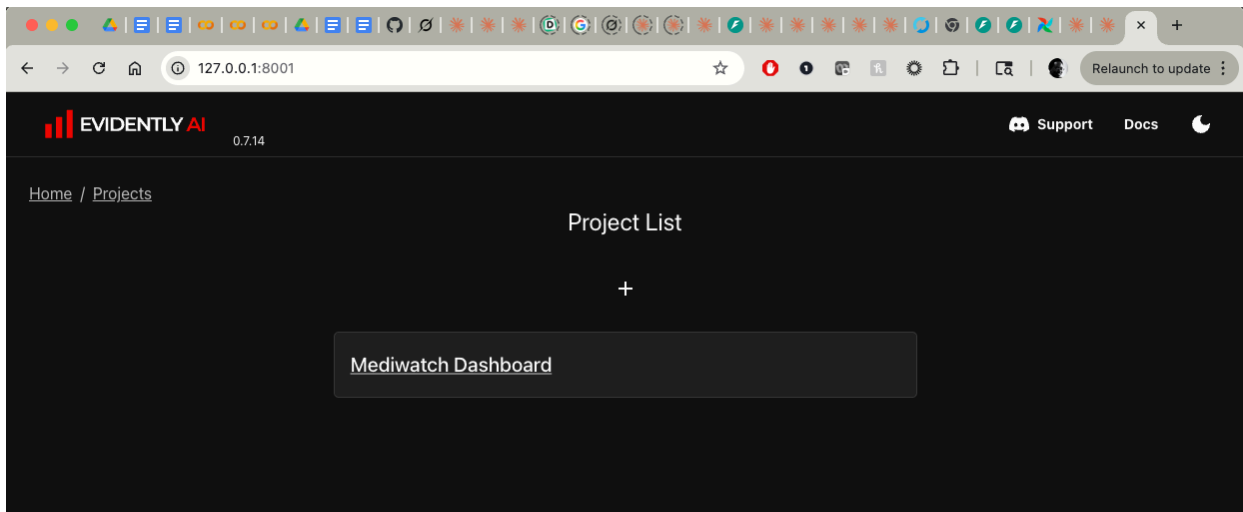
Evidently UI

Now, lets open the evidently UI

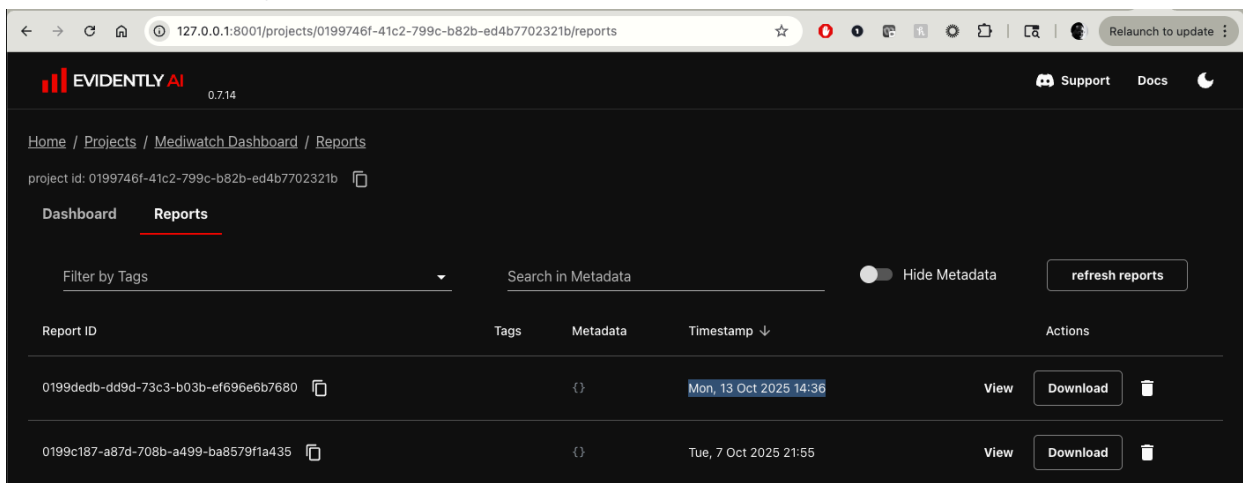
```
evidently ui --workspace
/home/victor/Documents/interview_kickstart/Capstone/Capstone-MLOPS-Mediawatch/ev
idently/evidently-data
```

```
23 }
PROBLEMS 4 OUTPUT DEBUG CONSOLE TERMINAL PORTS 4
● Capstone/Capstone-MLOPS-Mediwatch$ cd evidently/
(/home/victor/Documents/interview_kickstart/Capstone/Capstone-MLOPS-Mediwatch/env) victor@victor-Asus-PC:~/Documents/interview_kickstart/
○ Capstone/Capstone-MLOPS-Mediwatch/evidently$ evidently ui --workspace /home/victor/Documents/interview_kickstart/Capstone/Capstone-MLOPS-
Mediwatch/evidently/evidently-data
INFO: Started server process [3712844]
INFO: Waiting for application startup.
INFO: Application startup complete.
INFO: Uvicorn running on http://127.0.0.1:8000 (Press CTRL+C to quit)
INFO: 127.0.0.1:49260 - "GET / HTTP/1.1" 200 OK
INFO: 127.0.0.1:49260 - "GET /static/js/index-C1CmpqGk.js HTTP/1.1" 200 OK
INFO: 127.0.0.1:49260 - "GET /static/css/index-uUDKGmkJ.css HTTP/1.1" 200 OK
INFO: 127.0.0.1:49260 - "GET /api/version HTTP/1.1" 200 OK
Observer for '/home/victor/Documents/interview_kickstart/Capstone/Capstone-MLOPS-Mediwatch/evidently/evidently-data' started
INFO: 127.0.0.1:49246 - "GET /api/projects HTTP/1.1" 200 OK
INFO: 127.0.0.1:49268 - "GET /manifest.json HTTP/1.1" 200 OK
INFO: 127.0.0.1:49260 - "GET /favicon.ico HTTP/1.1" 200 OK
INFO: 127.0.0.1:56632 - "GET /api/projects/0199746f-41c2-799c-b82b-ed4b7702321b/info HTTP/1.1" 200 OK
INFO: 127.0.0.1:56648 - "GET /static/js/dashboard-main-C2fnAhD0.js HTTP/1.1" 200 OK
INFO: 127.0.0.1:56660 - "GET /static/js/Pie-D2vhsTcR.js HTTP/1.1" 200 OK
INFO: 127.0.0.1:56660 - "GET /api/v2/dashboards/0199746f-41c2-799c-b82b-ed4b7702321b HTTP/1.1" 200 OK
INFO: 127.0.0.1:56660 - "GET /api/projects/0199746f-41c2-799c-b82b-ed4b7702321b/snapshots HTTP/1.1" 200 OK
```

Here is the UI

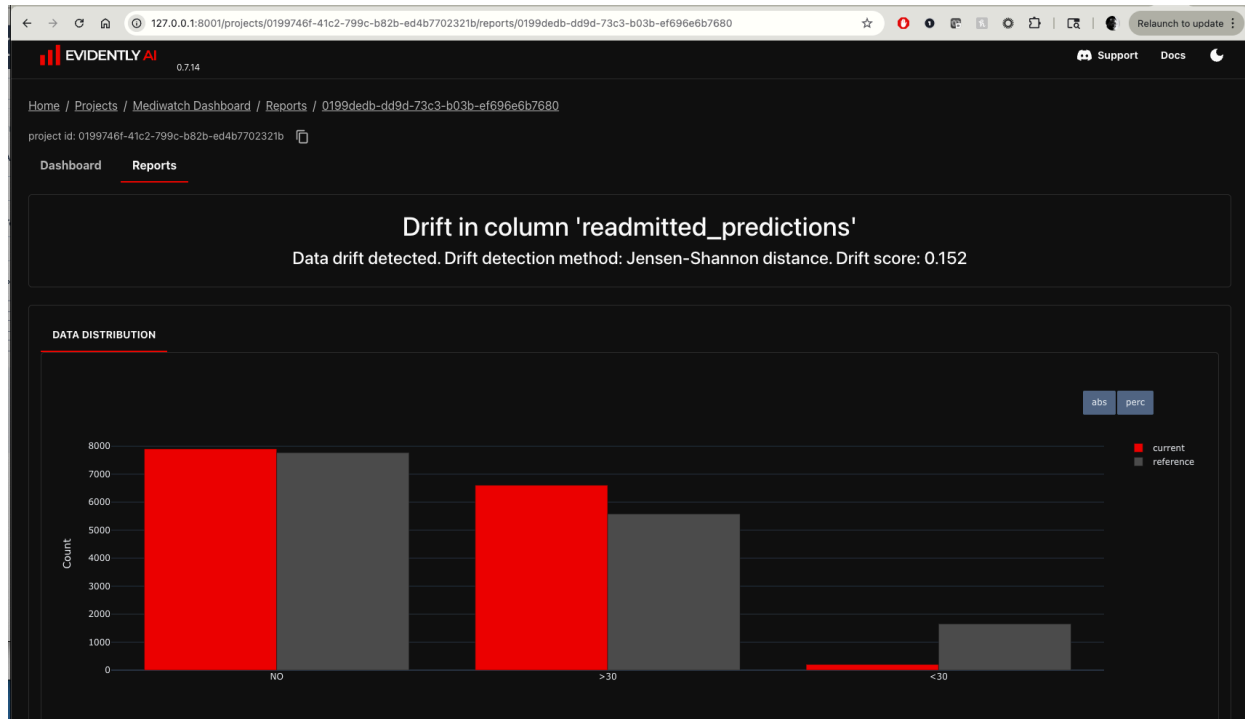


And when I look at my reports...



Let's look at the latest one:

Here is my Data Drift:



Continuous Integration, Deployment and Orchestration

Airflow

I chose to use airflow for my continuous integration and deployment pipeline. In my case look at **airflow/dags/training_api_dag.py**

Right at the bottom are the dags I am using to create the new docker image and deploy to kubernetes


```
airflow > dags > training_api_dag.py > ...
348 )
349
350 #Hit the build docker image endpoint to build the new docker image
351 build_docker_image = PythonOperator(
352     task_id='build_docker_image',
353     python_callable=call_dockerize_api_robust,
354     dag=dag,
355 )
356
357 update_deployment_yaml = PythonOperator(
358     task_id='update_deployment_yaml',
359     python_callable=create_deployment_yaml,
360     dag=dag
361 )
362
363 deploy_image_to_kubernetes = PythonOperator(
364     task_id='deploy_image_to_kubernetes',
365     python_callable=deploy_to_kubernetes,
366     dag=dag
367 )
368
369
370 # Set up task dependencies: api_health_check -> kubernetes_health_check -> training -> docker_build -> update_deployment_yaml
371 api_health_check >> kubernetes_health_check >> start_training_robust >> build_docker_image >> update_deployment_yaml >> deploy_image_to_kubernet
```

Scheduling

The 'training_and_docker_build' dag runs the complete pipeline.

The dag can be run on a whatever schedule you desire by modifying the **schedule_interval** below

```
22
23 # Define the combined DAG for training and Docker build
24 dag = DAG(
25     'training_and_docker_build',
26     default_args=default_args,
27     description='Train model and build Docker image',
28     schedule_interval='@daily',
29     #schedule_interval=None,
30     catchup=False,
31     tags=['api', 'training', 'ml', 'docker', 'build'],
32 )
33
34 # Call training endpoint
```

Start Airflow

Use docker compose up -d

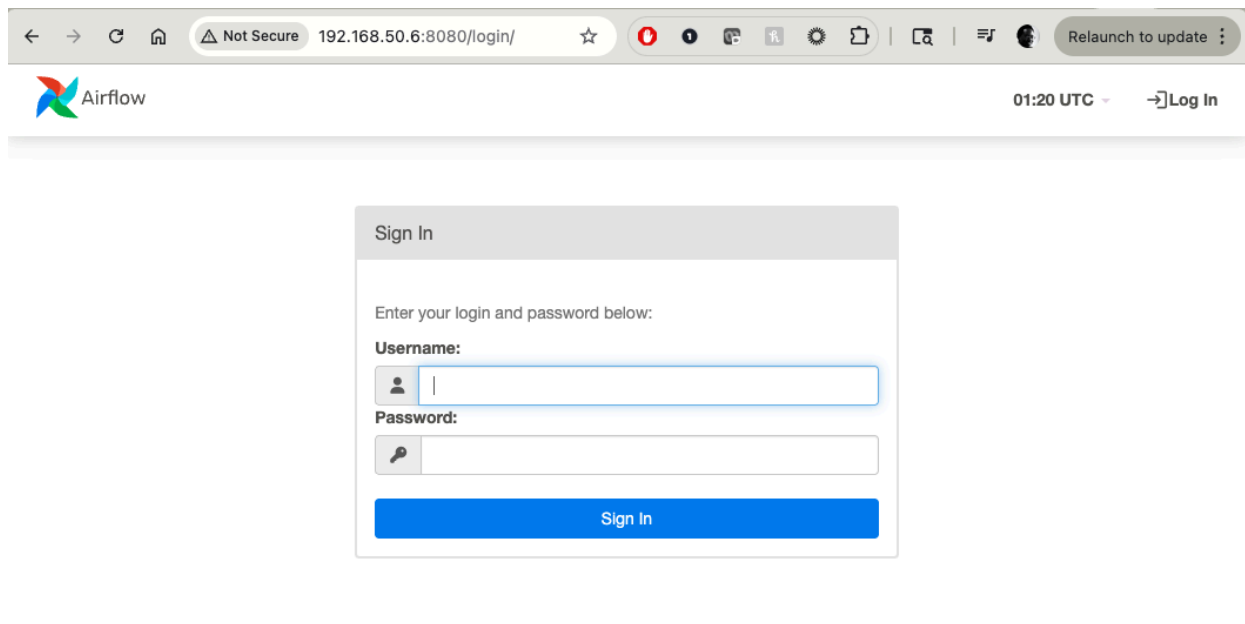
```
conda activate base
(base) victor@victor-Asus-PC:~/Documents/interview_kickstart/Capstone/Capstone-MLOPS-Mediwatch$ conda activate base
(base) victor@victor-Asus-PC:~/Documents/interview_kickstart/Capstone/Capstone-MLOPS-Mediwatch$ cd airflow/
(base) victor@victor-Asus-PC:~/Documents/interview_kickstart/Capstone/Capstone-MLOPS-Mediwatch/airflow$ docker compose up -d
WARN[0000] /home/victor/Documents/interview_kickstart/Capstone/Capstone-MLOPS-Mediwatch/airflow/docker-compose.yaml: the attribute `version` is obsolete, it will be ignored, please remove it to avoid potential confusion
[+] Running 6/8
  ✓ Network airflow_default          Created           0.0s
  ✓ Container airflow-redis-1        Waiting          5.5s
  ✓ Container airflow-postgres-1     Waiting          5.5s
  ✓ Container airflow-airflow-init-1 Created           0.1s
  ✓ Container airflow-airflow-scheduler-1 Created          0.0s
  ✓ Container airflow-airflow-webserver-1 Created          0.0s
  ✓ Container airflow-airflow-triggerer-1 Created          0.0s
  ✓ Container airflow-airflow-worker-1 Created          0.0s
```

...

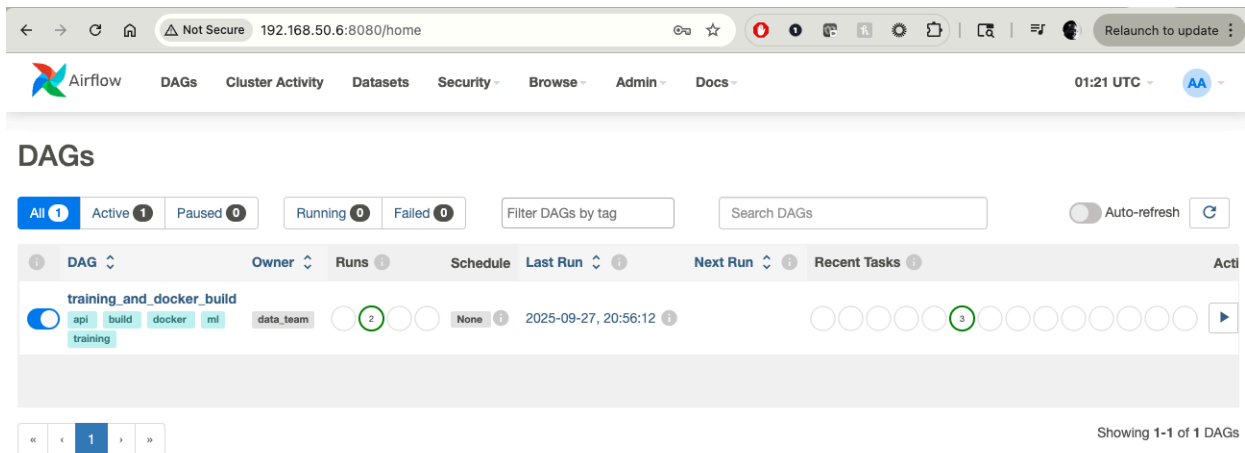
...

```
conda activate base
(base) victor@victor-Asus-PC:~/Documents/interview_kickstart/Capstone/Capstone-MLOPS-Mediwatch$ conda activate base
(base) victor@victor-Asus-PC:~/Documents/interview_kickstart/Capstone/Capstone-MLOPS-Mediwatch$ cd airflow/
(base) victor@victor-Asus-PC:~/Documents/interview_kickstart/Capstone/Capstone-MLOPS-Mediwatch/airflow$ docker compose up -d
WARN[0000] /home/victor/Documents/interview_kickstart/Capstone/Capstone-MLOPS-Mediwatch/airflow/docker-compose.yaml: the attribute `version` is obsolete, it will be ignored, please remove it to avoid potential confusion
[+] Running 8/8
  ✓ Network airflow_default          Created           0.0s
  ✓ Container airflow-redis-1        Healthy           6.8s
  ✓ Container airflow-postgres-1     Healthy           6.8s
  ✓ Container airflow-airflow-init-1 Exited           18.8s
  ✓ Container airflow-airflow-scheduler-1 Started           19.0s
  ✓ Container airflow-airflow-webserver-1 Started           18.9s
  ✓ Container airflow-airflow-triggerer-1 Started           19.0s
  ✓ Container airflow-airflow-worker-1 Started           18.9s
(base) victor@victor-Asus-PC:~/Documents/interview_kickstart/Capstone/Capstone-MLOPS-Mediwatch/airflow$
```

Note that the 'exited' init process above is ok. It does its initial work then goes away. Here is my login page. Note that on my setup, it usually takes ~ 10 s from docker compose up completion to having the login page appear



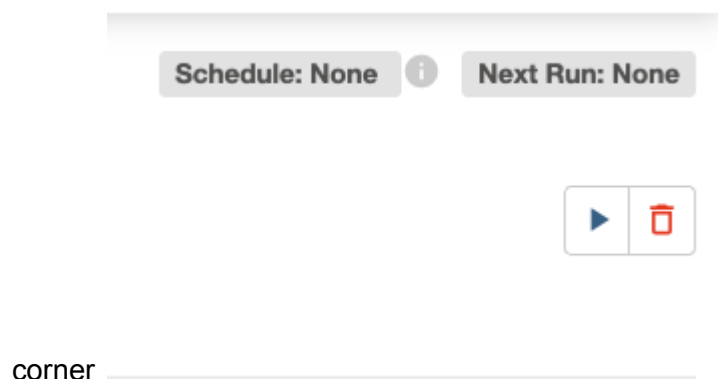
Login as **admin/admin123**



The screenshot shows the Apache Airflow web interface. The top navigation bar includes links for DAGs, Cluster Activity, Datasets, Security, Browse, Admin, and Docs. The main header displays the time as 01:21 UTC. The 'DAGs' section is active, showing a list of DAGs. The first DAG, 'training_and_docker_build', is highlighted. It has a status of 'Active' (indicated by a blue circle) and a 'Run' button (indicated by a green circle with a play icon). The 'Last Run' column shows the date and time '2025-09-27, 20:56:12'. The 'Recent Tasks' column shows a list of tasks with their respective statuses. The bottom of the interface shows a pagination bar with '1' selected and a message 'Showing 1-1 of 1 DAGs'.

Run the training_and_docker build dag

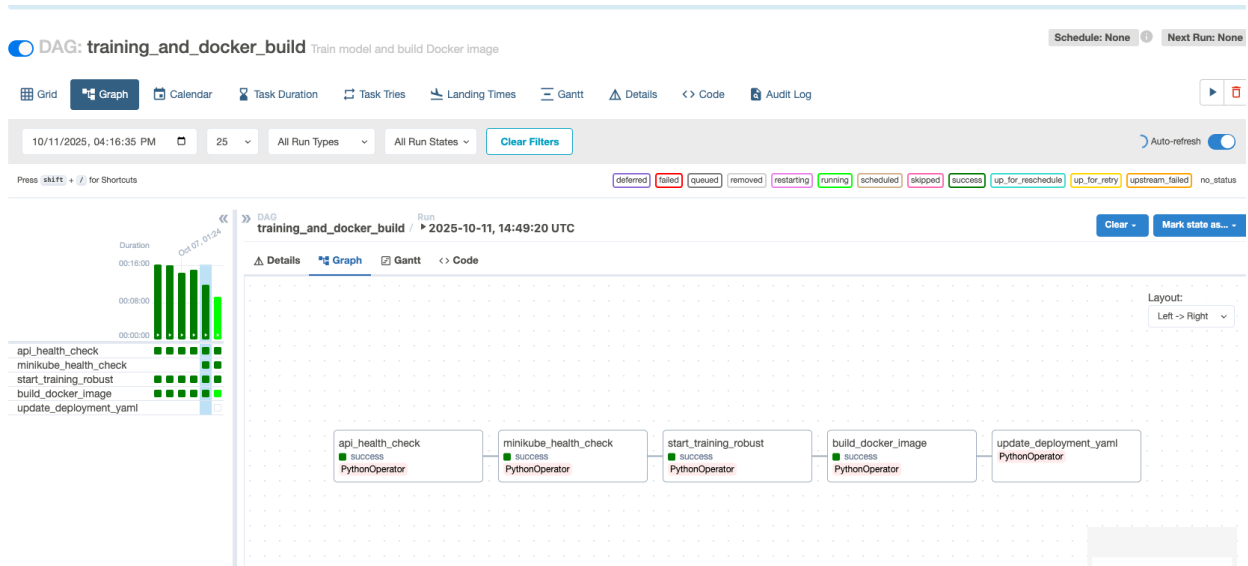
Click on the **'training_and_docker_build'** dag, then click on the run button in the top right



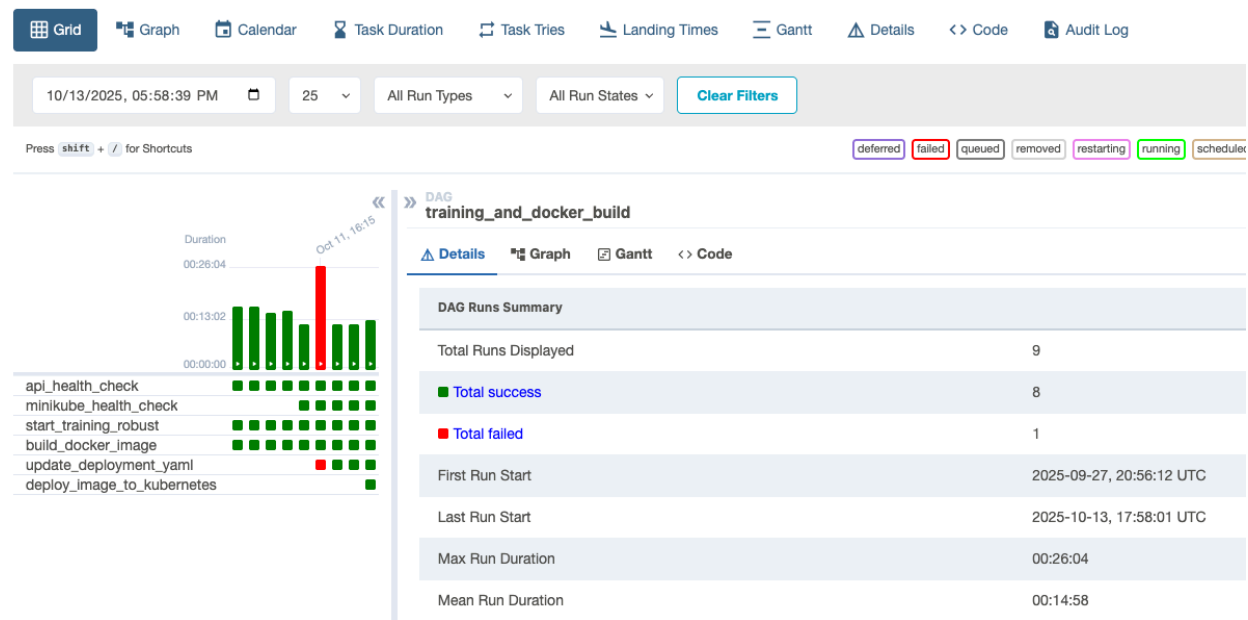
The screenshot shows the details page for the 'training_and_docker_build' DAG. The top section displays 'Schedule: None' and 'Next Run: None'. Below this, there are two buttons: a blue play button (Run) and a red trash can icon (Delete). The bottom of the interface shows a pagination bar with '1' selected and a message 'Showing 1-1 of 1 DAGs'.

As the dag runs, active items move from bright green (active) to dark green (completed)

And now build docker image is active



And now it is complete:



And here is the resulting docker image in minikube:

```
PROBLEMS 12 CPU 67 DEBUG CONSOLE TERMINAL PORTS 100
(base) victor@victor-Asus-PC:~/Documents/interview_kickstart/Capstone/Capstone-MLOPS-Mediawatch$
(base) victor@victor-Asus-PC:~/Documents/interview_kickstart/Capstone/Capstone-MLOPS-Mediawatch$ docker images
REPOSITORY          TAG          IMAGE ID       CREATED        SIZE
test_image1760378329 latest       e0df79810aab   9 minutes ago  3.78GB
test_image1760372126 latest       a722e9def9b6   2 hours ago    3.77GB
test_image1760369737 latest       1df71c62037b   3 hours ago    3.77GB
test_image_1000     latest       485cfd0b87d4   4 hours ago    3.77GB
registry.k8s.io/kube-controller-manager v1.33.1     ef43894fa110   5 months ago   94.6MB
registry.k8s.io/kube-apiserver          v1.33.1     c6ab243b29f8   5 months ago   102MB
registry.k8s.io/kube-scheduler          v1.33.1     398c985c0d95   5 months ago   73.4MB
registry.k8s.io/kube-proxy              v1.33.1     b79c189b052c   5 months ago   97.9MB
registry.k8s.io/etcd                    3.5.21-0    499038711c08   6 months ago   153MB
registry.k8s.io/coredns/coredns         v1.12.0     1cf5f116067c   10 months ago  70.1MB
registry.k8s.io/pause                   3.10        873ed7510279   16 months ago  736kB
kubernetesui/dashboard                 <none>       07655ddf2eeb   3 years ago    246MB
kubernetesui/metrics-scrapers           <none>       115053965e86   3 years ago    43.8MB
gcr.io/k8s-minikube/storage-provisioner v5          6e38f40d628d   4 years ago    31.5MB
(base) victor@victor-Asus-PC:~/Documents/interview_kickstart/Capstone/Capstone-MLOPS-Mediawatch$
```

Stop Airflow

Note: you can turn off airflow with docker compose down if you are done with it.

```
airflow-postgres-1
832e4654be96 redis:latest "docker-entrypoint.s..." 32 minutes ago Up 32 minutes (healthy) 6379/tcp
airflow-redis-1
(base) victor@victor-Asus-PC:~/Documents/interview_kickstart/Capstone/Capstone-MLOPS-Mediawatch/airflow$ docker compose down
WARN[0000] /home/victor/Documents/interview_kickstart/Capstone/Capstone-MLOPS-Mediawatch/airflow/docker-compose.yaml: the attribute 'version' is ob
solete, it will be ignored, please remove it to avoid potential confusion
[+] Running 8/8
  ✓ Container airflow-airflow-scheduler-1 Removed 3.0s
  ✓ Container airflow-airflow-worker-1 Removed 4.0s
  ✓ Container airflow-airflow-webserver-1 Removed 5.2s
  ✓ Container airflow-airflow-triggerer-1 Removed 2.5s
  ✓ Container airflow-airflow-init-1 Removed 0.2s
  ✓ Container airflow-postgres-1 Removed 0.2s
  ✓ Container airflow-redis-1 Removed 0.2s
  ✓ Network airflow_default Removed 0.2s
(base) victor@victor-Asus-PC:~/Documents/interview_kickstart/Capstone/Capstone-MLOPS-Mediawatch/airflow$
```